Flows & Liquidity

Deleveraging likely behind us

- **We find that investors deleveraging in the tech and semiconductor space, including memory stocks, has advanced faster than we had previously anticipated.**
- **As a result, we now see more limited room for any further deleveraging.**
- **The collapse of the odds of the Clarity Act passing by the Senate represents a setback for crypto markets.**

Cross Asset Fund Flow Monitor

Current level shows the latest percentile of weekly flows; Min is denoted by 0 and Max by 1. As of 22nd July 26.

MF & ETF Flows	Min	Max	4 wk avg ($bn)	2025 avg ($bn)
All Equities			12.0	8.1
All Bonds			17.8	11.3
US Equities			-2.1	3.5
US Bonds			9.5	4.3
Non-US Equities			14.1	4.6
Non-US Bonds			8.3	7.0
US HG Bonds			11.8	3.4
US HY Bonds			0.2	0.4
US Lev. Loans *			0.3	0.0
US MMFs			0.5	4.0
EM Equities			2.5	0.9
EM Bonds			0.93	0.50
Japan Equities			2.3	-0.1
China Equities			-0.99	-0.15
Europe				
Europe Equities			0.2	0.9
Europe Bonds			5.9	4.3
Europe HG Bonds			0.6	0.9
Europe HY Bonds			0.27	0.17
Europe MMFs			1.6	3.8
Other Equities			9.18	3.00

Source: Lipper, ICI, Bloomberg Finance L.P. and J.P. Morgan Flows & Liquidity.
* US LL historical flows are monthly averages converted to weekly for comparison. China onshore A-share ETFs have been excluded.

Global Markets Strategy

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Cross Asset Positioning Monitor

Current level shows the latest percentile, Min is denoted by 0 and Max by 1.

As of 28-Jul-26	MIN	MAX	Current percentile
Equities			0.59
Govt Bonds			0.64
Credit			0.27
Dollar			0.81
Commodities ex Gold			0.64
Gold			0.54
Bitcoin			0.55
EM Equities			0.69
EM Bonds/FX			0.24
Japan Equities			0.60
Europe Equities			0.88
US Equity Sectors:			
Energy			0.73
Materials			0.57
Industrials			0.65
Discretionary			0.65
Staples			0.27
Health Care			0.48
Financials			0.25
Technology			0.56
Communication Services			0.32
Utilities			0.54

Source: J.P. Morgan Flows & Liquidity.
Cross Asset Positioning Monitor aggregates across the various position indicators of Appendix ranging from positioning proxies across various futures contracts, momentum signals as proxies of how trend-following funds/CTAs are positioned, mutual fund betas as proxies of how mutual fund managers are positioned, risk parity fund positioning and leverage proxies, hedge fund betas as proxies of how hedge fund managers are positioned, client surveys, asset allocation estimates of private non-bank investors at global level, short interest indicators, etc.

See page 26 for analyst certification and important disclosures.

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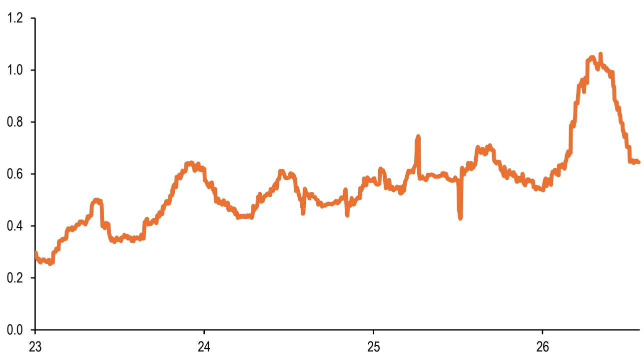
- Two weeks ago, we argued that the deleveraging phase that started in June appeared to be still ongoing and saw room for further deleveraging by investors. Two weeks later, we find that this deleveraging has advanced faster than we had anticipated. As a result, we now see more limited room for any further deleveraging.
- Where has deleveraging advanced:
  - 1) Starting with hedge funds, we had already noted signs of deleveraging in our previous publication two weeks ago as daily reporting Equity Long/Short funds exhibited less close relationship with semiconductor stocks. By updating our leverage proxy for daily reporting Equity Long/Short hedge funds with respect to the semiconductor (SOX) index, we find evidence that the previous increase in leverage by equity focused hedge funds during April/May has been largely unwound (Figure 1).

Figure 1: Our leverage proxy based on volatility ratios for daily reporting Equity Long/Short hedge funds with respect to the semiconductor (SOX) index

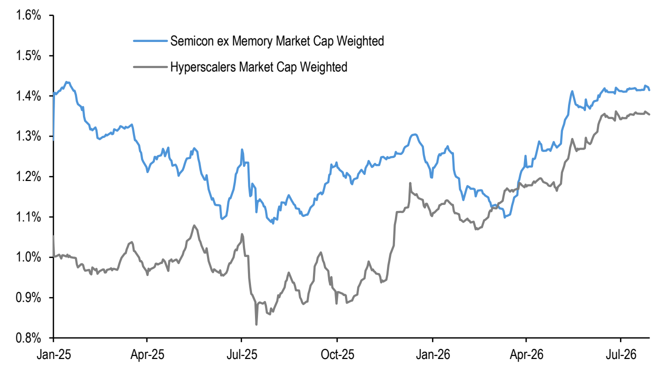


Source: HFR, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

  - 2) The elevated short interest on individual tech stocks (Figure 2) or important semiconductor ETFs such as SMH and DRAM (Figure 3) is in our opinion indicative of low net exposure to tech stocks by hedge funds.

Figure 2: Short interest on Semiconductors ex Memory & Hyperscalers individual stocks

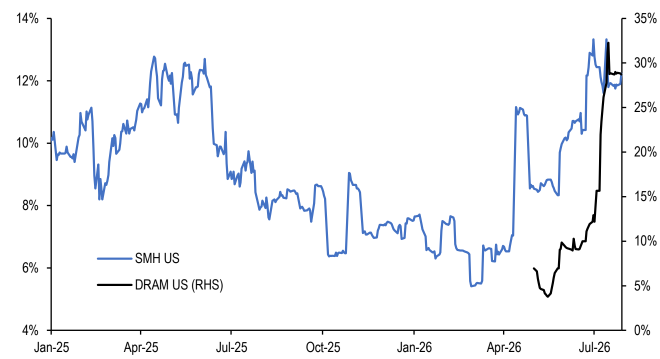
Short Interest as a % share of shares outstanding.



Source: S3, J.P. Morgan Flows & Liquidity.

Figure 3: Short interest on SMH and DRAM US Equity ETFs

Short Interest as a % share of shares outstanding.



Source: S3, J.P. Morgan Flows & Liquidity.

  - 3) The collapse of momentum signals in tech heavy equity indices suggests that momentum traders such as CTAs have largely unwound previous long equity positions in Nasdaq, Kospi, Taiwan and Nikkei (Figure 4 and Figure 5). In addition, Figure 5 suggests that the previously crowded relative trade of selling China's internet companies (HSCEI Index) and buying Korea/Taiwan/Japan equities has reversed fully.

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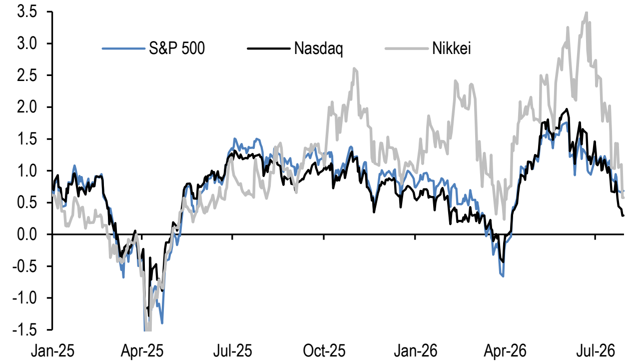
Figure 4: Momentum signals for major equity indices

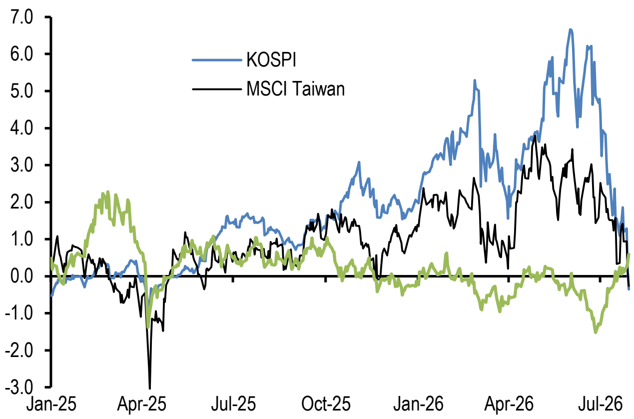
Average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 below in the Appendix.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Figure 5: Momentum signals for Asian equity indices

Average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 below in the Appendix.

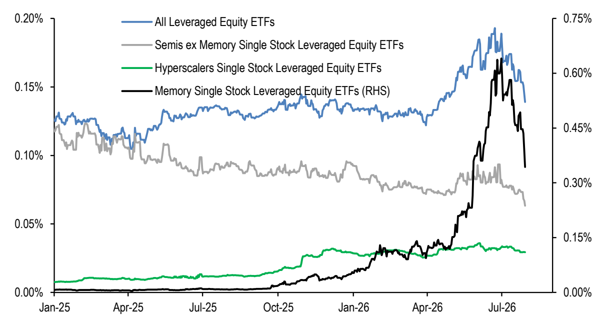


Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- 4) For leveraged equity ETFs, we had previously argued that volatile range-trading would suffice to naturally reduce their AUM due to convexity. For example, if the underlying index falls 10% one day and rises 11.1% the following day to reach its previous level, a 3x leveraged ETF would be down 30% on the first day and up 33.3% in the second day, thus ending 7% lower from its previous level. In other words, the excessive growth of leveraged equity ETFs can be self-correcting due to convexity: if excess leverage creates higher frequency of VaR shocks, a volatile range-trading in equities would over time shrink their AUM. And the price effects due to convexity would dominate over continued capital inflows into these leveraged ETFs.

- Instead of range-trading, we got steep price declines in semiconductor and memory stocks over the past two weeks. As a result, the reduction in the AUM of leveraged equity ETFs has taken place much faster than we previously envisaged. In fact, after peaking in June, the AUM ratio of leveraged equity ETFs to the market cap of underlying stocks has largely normalized for all leveraged equity ETFs (including both equity index and individual stock leveraged ETFs), unwinding around 85% of its previous April/May increase (blue line in Figure 6). This is also true for semiconductor stocks ex memory (grey line in Figure 6). For memory stocks, the ratio has unwound around 55% of its previous increase during April/May (black line in Figure 6). In other words, the leveraged ETF problem as a cause of deleveraging and source of excessive volatility should no longer be a significant issue for equities overall and should be much less of an issue for individual memory stocks going forward.

Figure 6: AUM of leveraged equity ETFs as % of the market cap of underlying stocks

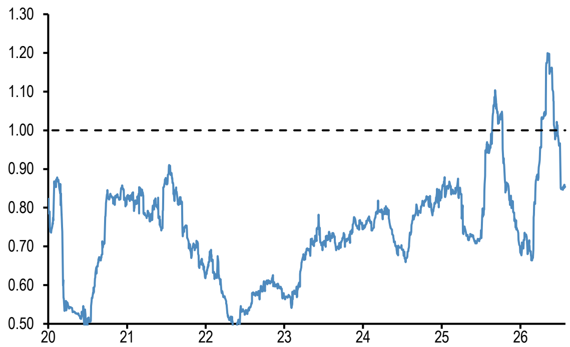


Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- 5) In the risk parity fund space, our leverage proxy has already normalized as shown in Figure 7, so excessive leverage by risk parity funds should no longer present a significant headwind for equities going forward.

Figure 7: Implied leverage of risk-parity funds

Ratio of 3-month rolling volatility of risk-parity funds' returns to the volatility of our risk parity strategy benchmark.

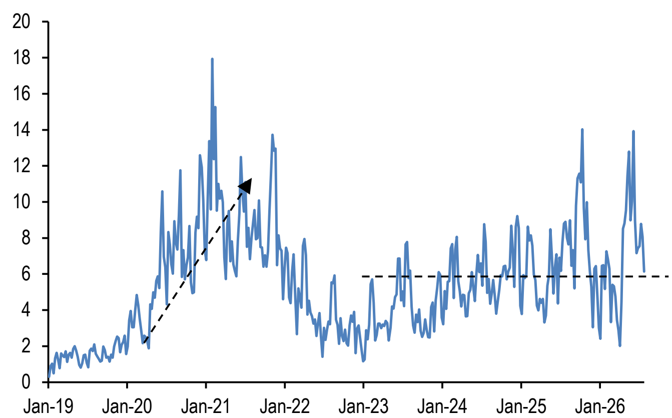


Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- 6) Similarly, in the options space, previous extreme call option buying by retail investors appears to have normalized to historical averages (Figure 8). As a reminder, our proxy of retail investors' option buying based on OCC data (on exchange-traded call option buying by customers with less than 10 contracts), retreated significantly after peaking on June 5th. The peak on June 5th at close to 14 million contracts had matched the previous peaks seen in October 2025 and November 2021.

Figure 8: Exchange-traded Call Option Buys at Open minus Sells at Open for Customers with less than 10 contracts for options on individual equities

In mn contracts. Last obs is for the week ending 24th July 2026.

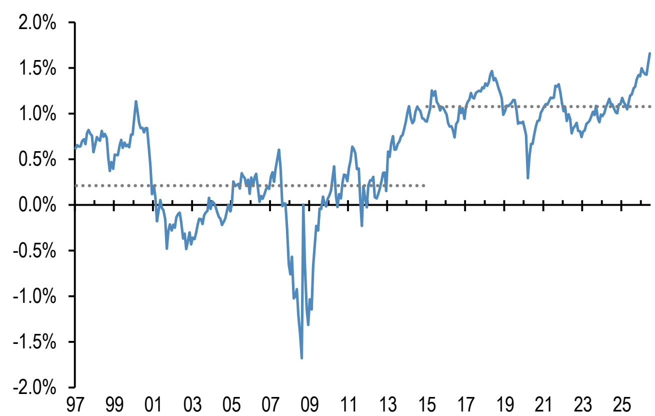


Source: OCC, J.P. Morgan Flows & Liquidity.

- 7) Leverage in margin accounts stood at very elevated level in June as shown in Figure 9 and thus significantly more deleveraging would need to be seen in July to be able to say that margin account leverage no longer presents a significant headwind for equities. As a reminder, for margin accounts we use the NYSE Net Debit balance as a proxy for US individual investor leverage. While in principle it includes both retail and institutional investors, we believe the majority is likely retail investors. In contrast to hedge funds, who can get leverage more easily or cheaply via options and futures, individuals rely more on Federal Reserve Regulation T. This regulation, which governs customer cash accounts and the amount of credit that brokerage firms and dealers may extend to customers for the purchase of securities, stipulates that one can borrow up to 50% of the purchase price of securities that can be purchased on margin. The NYSE Net Debit balance is calculated as the difference between margin debit balance minus the sum of total credit balances (cash accounts credit + margin account credit). This leverage proxy suggests US individual investor leverage via margin accounts stood at very elevated levels in June and we suspect that a significant decline has taken place in July. Unfortunately, we need to wait for a month to get confirmation of a potential decline in July, as the July observation will be reported at the end of August.

Figure 9: Net Debit balances in NYSE margin accounts

The NYSE margin account Net Debit balance is equal to the margin debit balance minus the sum of total credit balances (cash account credit +margin account credit). The blue line is constructed by the Net Debit balance in $bn divided by the market capitalisation of the S&P500 Index. Last monthly observation is for June 2026.



Source: FINRA, J.P. Morgan Flows & Liquidity.

- In all, we find that investors deleveraging in the tech and semiconductor space, including memory stocks, has advanced faster than we had previously anticipated. As a result, we now see more limited room for any further deleveraging. For Korea in particular, we recommend the recent note by our colleague Mixo Das "Korea Equity Strategy – Update on the de-leveraging process in Korea (2)" July 29th 2026, who arrived at a similar conclusion to ours.

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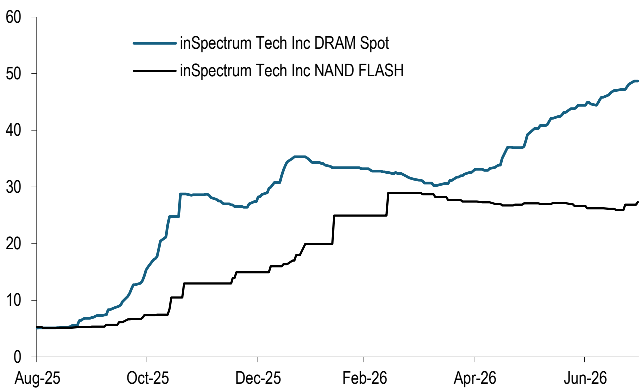
- Finally, assuming our thesis above proves correct and most of the investor deleveraging is behind us, we believe the equity market will likely receive support from the following developments:
  - a) Memory prices are still rising, providing fundamental support to memory makers which have been at the center of recent deleveraging (Figure 10).

Figure 10: inSpectrum Tech memory prices



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

  - b) The hyperscalers capex projections have been raised significantly by analysts for both 2026 and 2027 (Figure 11) to $770bn and $1tr from $758bn and $925bn on July 1st, respectively.

Figure 11: US hyperscalers capex projections by the consensus of bottom-up analysts as reported by Bloomberg

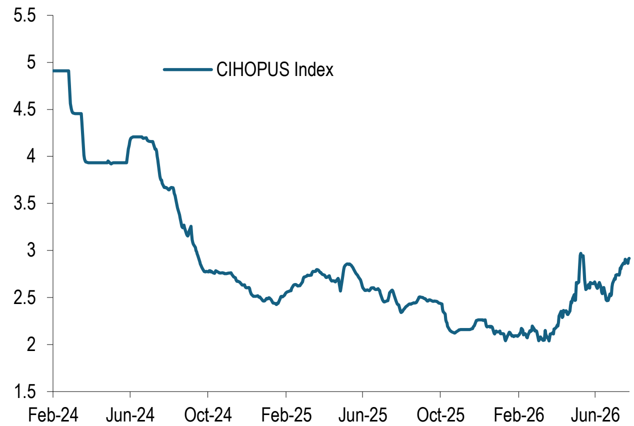
In $bn	23	24	25	26E	27E	28E	29E	30E
Google	32.3	52.5	91.4	197.0	290.2	312.7	305.4	315.5
Amazon	52.7	83.0	131.8	200.1	238.7	250.7	252.0	256.1
Meta	27.3	37.3	69.7	135.6	175.0	194.8	192.0	196.0
Microsoft	28.1	44.5	64.6	160.6	198.1	223.8	255.9	279.6
Oracle	8.7	6.9	21.2	76.7	99.3	102.7	90.5	73.9
Total	149.0	224.1	378.7	770.1	1,001.4	1,084.7	1,095.8	1,121.2
Y/Y growth	-4%	50%	69%	103%	30%	8%	1%	2%

Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

  - c) The price of AI compute is holding up as proxied by Hopper rental prices in Figure 12. The price for AI compute would be key for the ability of hyperscalers to monetize their AI capital spending. The higher the price for compute, the higher the ability by hyperscalers to maintain or increase their profit margins. After some persistent downward pressure on compute pricing into end-2025, there have been some signs of improvement in recent months. In addition, we find Figure 12 casts doubt to excessive obsolescence/ depreciation assumptions made by some investors.

Figure 12: Compute Desk's Hopper US Index

Aggregates listed on-demand and reserved prices for renting NVIDIA Hopper (H100 and H200) GPUs from US neocloud providers. The Index is priced in US dollars per GPU per hour.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

The collapse of the odds of the Clarity Act passing by the Senate represents a setback for crypto markets

- With the Senate prioritizing other legislation ahead of the summer recess, the prediction market odds of the Clarity Act passing before year-end collapsed to 37%, the lowest implied probability so far this year.
- The ethics issue has emerged as one of the biggest obstacles. Democrats have drawn a line on state enforcement and Republicans and the White House have drawn a line on DoJ enforcement alone. Negotiations are continuing, but it is unclear how this or other impasses on yield, DeFi, and illicit finance, that have not moved in six months, will be eventually resolved.
- We had previously argued (see F&L Feb 26th 2026) that the Market Structure Bill (i.e. Clarity Act) would act as positive catalyst for crypto markets for several reasons: the legislation introduces a clear framework for classifying crypto tokens as either digital commodities to be overseen by the CFTC or digital securities to be overseen by the SEC, the legislation provides a grace period for new projects to build and decentralize by permitting up to $75mn in annual capital raising without full SEC registration , the legislation provides a pathway for tokens initially offered as securities to be reclassified as digital commodities in secondary markets once they are sufficiently decentralized and the issuer no longer exercises a managerial role, the legislation establishes a comprehensive regulatory framework for crypto intermediaries by requiring registration with either the CFTC or SEC, ending regulation by enforcement, the legislation promotes the tokenization of traditional securities and real world assets as it makes it easier for them to be brought on-chain, the legislation allows miners, validators, wallet and protocol developers as pure software developers and to be exempted from “broker”/financial intermediary reporting as long as they do not engage in custodial activities or act as traditional financial middlemen.
- If the legislation is eventually passed, the above would together support the development of more institutional-grade market infrastructure, would ease current constraints around DeFi and stablecoin issuers, would increase onshore liquidity and trading volumes as activity gradually migrates from offshore venues to U.S.-compliant markets, and would lower the barriers for well-capitalized brokerages, exchanges, market makers, custodians, and bank-affiliated platforms to enter crypto markets.
- In fact, recent activity suggests that some of these positives are already materialising. Citadel Securities acquired a $400mn equity stake in Crypto.com. And the CFTC’s approval on May 29 of the first U.S.-regulated crypto perpetual futures products marked an important milestone, supporting the potential migration of perpetual futures liquidity from offshore markets back onshore.
- That said, we believe that there are certain aspects of the legislations in its current draft that could deter rather than promote institutional participation in crypto markets. One example is if DeFi is able to trade tokenized securities and tokenized derivatives entirely outside of SEC or CFTC jurisdiction, which the bill as currently drafted would allow. A second example is if crypto entities have to do little if any AML for performing the same activities as banks and broker dealers.
- Overall, the collapse of the odds of the Clarity Act passing by the Senate represents a setback for crypto markets. The longer the approval of the Clarity Act is postponed, the greater the threat to crypto markets from the growth of tokenization and blockchain-based applications eventually being absorbed by incumbent market infrastructure rather than accruing to public crypto networks (see F&L July 8th 2026).

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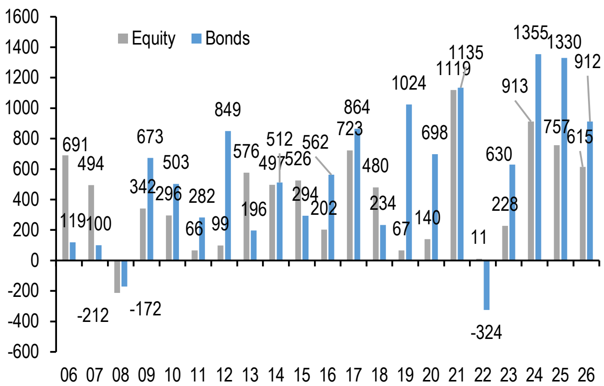
Appendix

Chart A1a: Global equity & bond fund flows

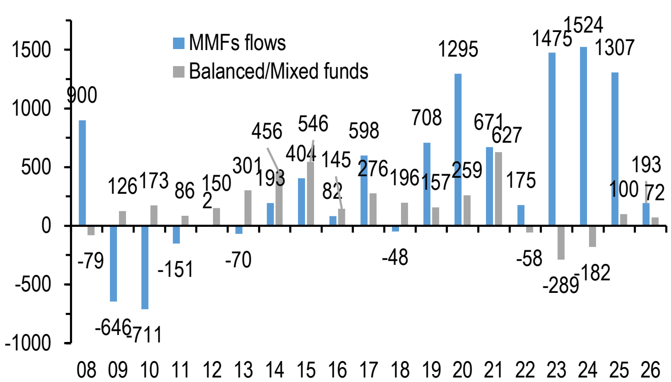
$bn per year of Net Sales, i.e. includes net new sales + reinvested dividends for Mutual Funds and ETFs globally, i.e. for funds domiciled both inside and outside the US. Flows come from ICI (worldwide data up to Q1’26). Data since then are a combination of monthly and weekly data from Lipper and Bloomberg.



Source: ICI, LSEG Lipper, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A1b: Quarterly Balanced/Mixed funds & MMFs flows worldwide

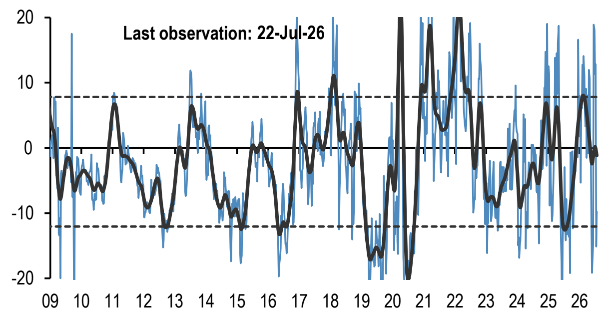
$bn per quarter of Net Sales, i.e. includes net new sales + reinvested dividends for Mutual Funds and ETFs globally, i.e. for funds domiciled both inside and outside the US. Data come from ICI (worldwide data) and are till Q1’26.



Source: ICI, J.P. Morgan Flows & Liquidity.

Chart A2: Fund flow indicator

Difference between flows into Equity and Bond funds: $bn per week. Difference between flows into Equity vs. Bond funds in $bn per week. Flows include Mutual Fund and ETF flows globally, i.e. funds domiciled both inside and outside the US (source: Lipper). The thin blue line shows the 4-week average of difference between Equity and Bond fund flows. Dotted lines depict ±1 StDev of the blue line since 2009. The thick black line shows a smoothed version of the same series. The smoothing is done using a Hodrick-Prescott filter with a Lambda parameter of 100. China onshore A-share ETFs have been excluded.



Source: LSEG Lipper, J.P. Morgan Flows & Liquidity.

Table A2: Trading turnover monitor

Volumes are monthly and Turnover ratio is annualised (monthly trading volume annualised divided by the amount outstanding). UST Cash is primary dealer transactions in all US government securities. UST futures are from Bloomberg Finance L.P. JGBs are OTC volumes in all Japanese government securities. Bunds, Gold, Oil and Copper are futures. Gold includes Gold ETFs. Min-Max chart is based on Turnover ratio. For Bunds and Commodities, futures trading volumes are used while the outstanding amount is proxied by open interest. The diamond reflects the latest turnover observation. The thin blue line marks the distance between the min and max for the complete time series since Jan-2005 onwards. Y/Y change is change in YTD notional volumes over the same period last year.

As of Jun-26	MIN	MAX	Turnover ratio	Vol (tr)	y/y chng
Equities					
EM Equity			1.4	$2.1	135%
DM Equity			1.4	$16.2	44%
Govt Bonds					
UST cash			12.3	$17.7	5%
UST futures			0.8	$14.9	8%
JGBs			49.1	¥5,035	11%
Bund futures			1.5	€10.1	18%
Credit					
US HG			1.1	$0.8	15%
US HY			0.5	$0.1	-1%
US Convertibles			3.6	$0.04	15%
Commodities					
Gold			52.4	$1.4	39%
Oil			80.1	$2.3	0%
Copper			1.9	$0.7	70%
Digital Assets					
CME Bitcoin			115.0	$0.061	-34%
CME Ethereum			134.8	$0.023	25%

* Data with one month lag

Source: Bloomberg Finance L.P., Federal Reserve, Trace, Japan Securities Dealer Association, WFE, J.P. Morgan Flows & Liquidity.

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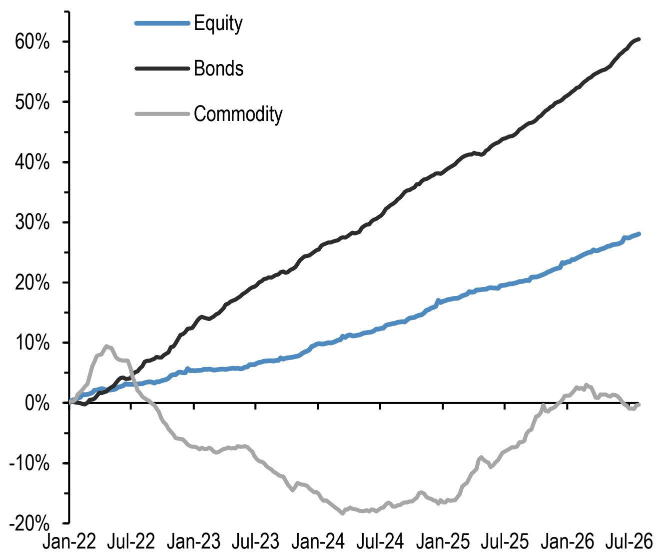
ETF Flow Monitor (as of 29th July)

Chart A3: Global Cross Asset ETF Flows

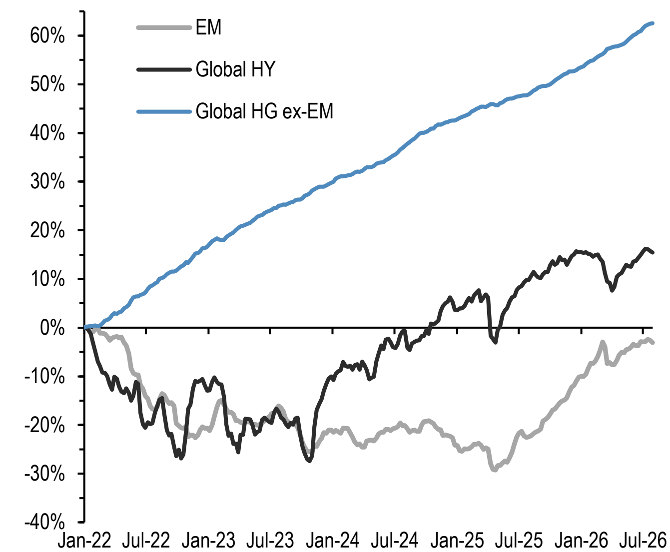
Cumulative flows into ETFs as a % of AUM



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity. China onshore A-share ETFs have been excluded.

Chart A4: Bond ETF Flows

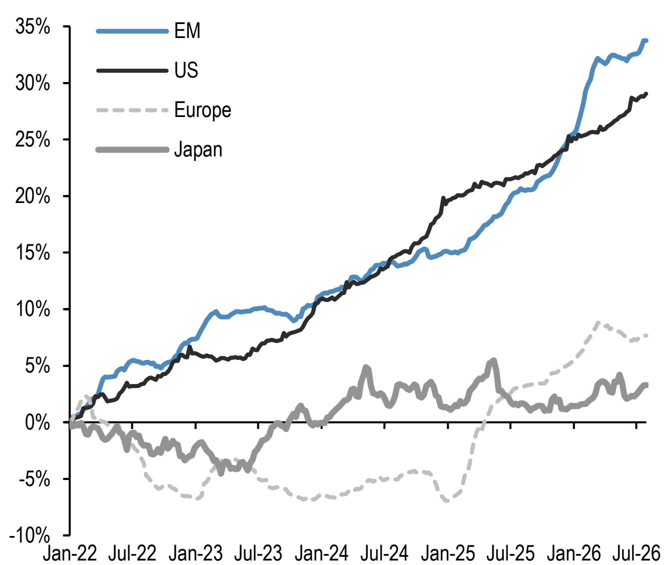
Cumulative flows into bond ETFs as a % of AUM



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A5: Global Equity ETF Flows

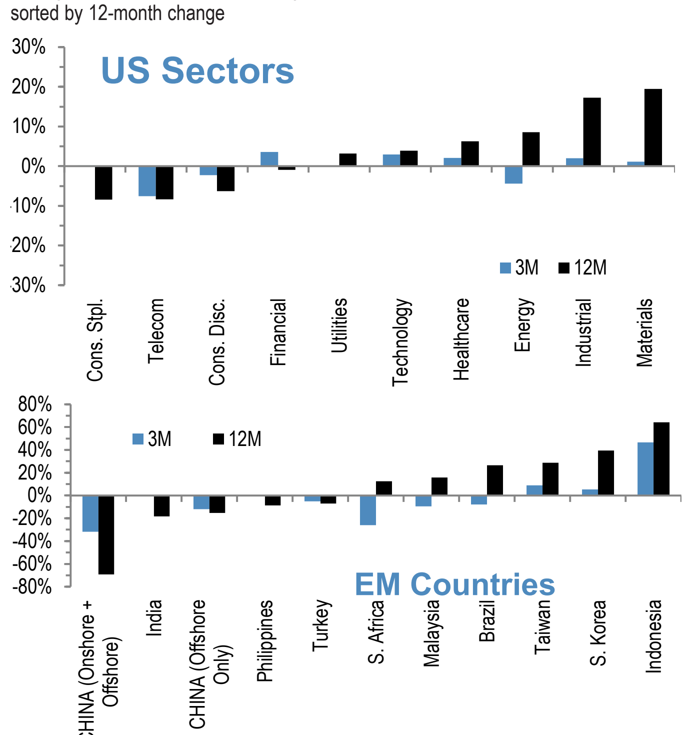
Cumulative flows into global equity ETFs as a % of AUM



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.
Note: We include ETFs with AUM > $200mn in all the flow monitor charts. In Chart A5 for Japan, we subtract the BoJ buying of ETFs. China onshore A-share ETFs have been excluded.

Chart A6a: Equity Sectoral and Regional ETF Flows

Rolling 3-month and 12-month change in cumulative flows as a % of AUM. Both sorted by 12-month change



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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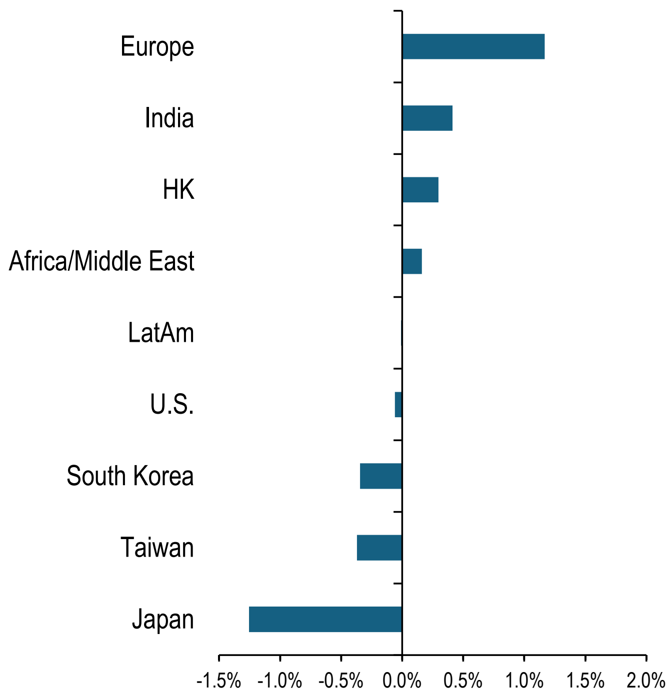
Chart A6b: Regional Equity Allocation Monitor Implied by ETFs

In %.



Source: Bloomberg Finance L.P., MSCI and J.P. Morgan Flows & Liquidity.
Note: Our regional equity allocation monitor implied by ETFs is based on a comparison of the share of each region in the equity ETF space relative to their corresponding shares in global equity indices.

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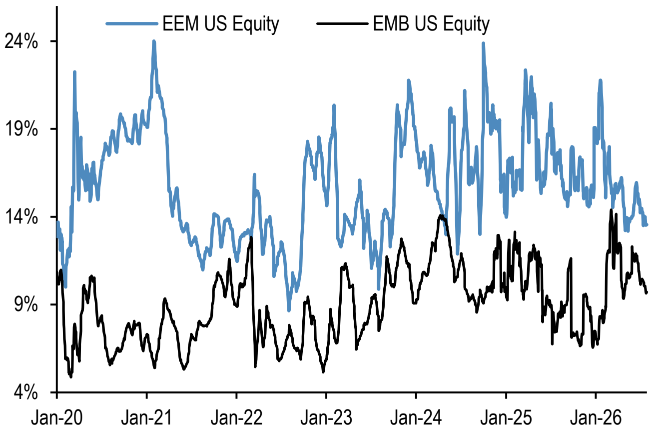
Short Interest Monitor

Chart A7: Short interest on the EEM and EMB US ETF

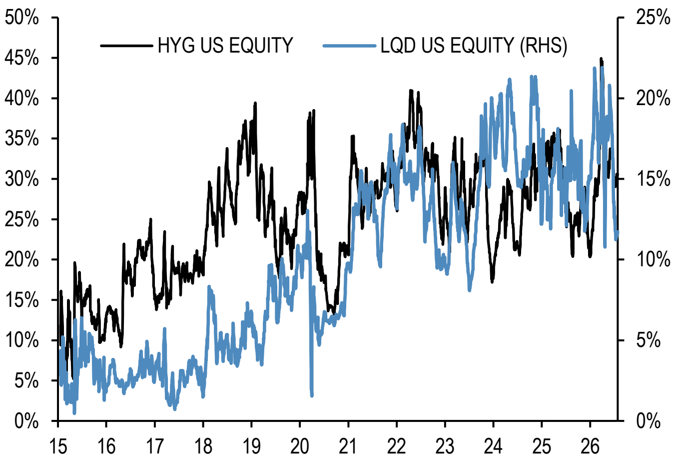
Short Interest as a % share of shares outstanding.



Source: S3, J.P. Morgan Flows & Liquidity.

Chart A8: Short interest on the LQD and HYG US ETF

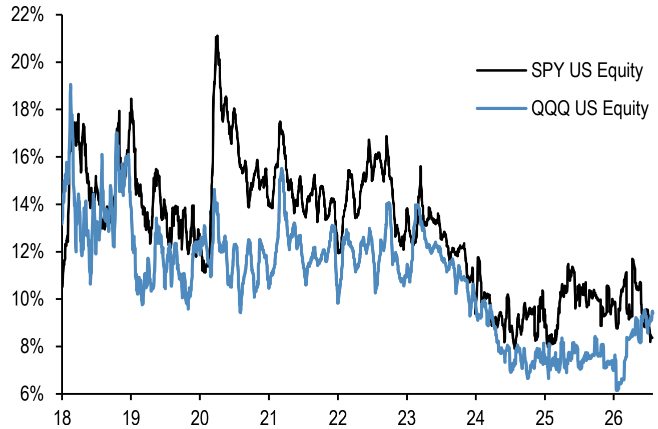
Short Interest as a % share of shares outstanding.



Source: S3, J.P. Morgan Flows & Liquidity.

Chart A9: Short interest on the SPY and QQQ US ETF

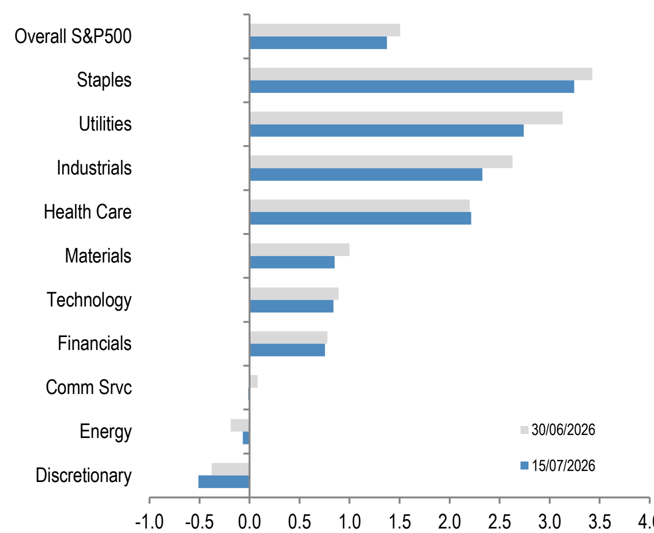
Short Interest as a % share of shares outstanding. Last obs is for 3rd July 2026.



Source: S3, J.P. Morgan Flows & Liquidity.

Chart A10: S&P500 sector short interest

Short interest as a % of shares outstanding based on z-scores. A strategy which overweights the S&P500 sectors with the highest short interest z-score (as % of shares o/s) vs. those with the lowest, produced an information ratio of 0.7 with a success rate of 56% (see F&L, Jun 28,2013 for more details).



Source: NYSE, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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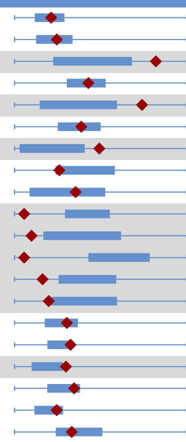
Chart A11a: Cross Asset Volatility Monitor 3m ATM Implied Volatility (1y history), as of 27th July-2026

This table shows the richness/cheapness of current three-month implied volatility levels (red dot) against their one-year historical range (thin blue bar) and the ratio to current realised volatility. Assets with implied volatility outside their 25th/75th percentile range (thick blue bar) are highlighted. The implied-to-realised volatility ratio uses 3-month implied volatilities and 1-month (around 21 trading days) realised volatilities for each asset.

Asset	Current	Low	Low date	High	High date	Upside	Downside	Implied/realized volatility
S&P 500	16%	13%	28-Aug-25	24%	27-Mar-26	8%	2%	1.57x
EuroSTOXX	16%	13%	18-Sep-25	26%	27-Mar-26	10%	3%	1.10x
Nikkei 225	31%	18%	29-Jul-25	33%	31-Mar-26	3%	12%	0.90x
Hang Seng	20%	17%	23-Dec-25	24%	26-Mar-26	4%	3%	1.04x
MSCI EM	31%	12%	15-Aug-25	38%	01-Jul-26	7%	19%	1.05x
Gold	21%	14%	19-Aug-25	33%	26-Mar-26	12%	7%	0.98x
Oil (brent)	57%	27%	19-Sep-25	89%	30-Mar-26	32%	31%	0.88x
Copper	20%	14%	18-Aug-25	37%	03-Feb-26	17%	6%	0.86x
BB commodity index	17%	13%	10-Dec-25	25%	30-Mar-26	8%	4%	1.05x
EUR/USD	8%	8%	16-Jul-26	13%	27-Mar-26	4%	0%	2.40x
USD/NOK	8%	8%	17-Dec-25	11%	30-Mar-26	3%	0%	1.37x
USD/JPY	7%	7%	16-Jul-26	10%	04-Aug-25	3%	0%	1.27x
GBP/USD	6%	6%	10-Jul-26	8%	27-Mar-26	2%	0%	1.35x
USD/CHF	7%	6%	29-May-26	9%	29-Jan-26	2%	1%	1.29x
10y US swaps	7%	6%	29-May-26	10%	27-Mar-26	3%	1%	1.06x
10y Eur swaps	60%	43%	26-Feb-26	97%	27-Mar-26	37%	17%	1.25x
CDX IG	46%	38%	29-Jul-25	66%	27-Mar-26	20%	8%	2.14x
CDX HY	40%	31%	09-Jan-26	57%	30-Mar-26	17%	9%	1.57x
iTraxx	45%	37%	09-Jan-26	70%	27-Mar-26	25%	8%	1.72x
iTraxx X/O	21%	19%	23-Dec-25	26%	17-Oct-25	4%	2%	0.81x



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Note: Swaps volatility is 3m 10y payer ATMF implied annualized BP vol and credit volatility is 3m 5y on-the-run ATM spread volatility. MSCI EM, Gold, Oil, Copper, BB Commodity Index and Treasury futures are 3m implied vol from Bloomberg.

Definitions:

Current:	Latest available closing level (27-Jul-26)
Low:	Lowest closing level in the last 1y
Low date:	Date the lowest closing level was reached (or the first time it was reached in the case of several identical low closing levels)
High:	Highest closing level in the last 1y
High date:	Date the highest closing level was reached (or the first time it was reached in the case of several identical high closing levels)
Graph:	Shows the current level and the 25th/75th percentile relative to the 1y high/low
Upside:	Implied return/volatility percentage points from current level up to the High (note: return is calculated as simple difference for spread products)
Upside (σ):	Upside in terms of standard deviations (Upside / Current 1y realized volatility)
Downside:	Implied return/volatility percentage points from current level down to the Low (note: return calculated as simple difference for spread products)
Downside (σ):	Downside in terms of standard deviations (Downside / Current 1y realized volatility)
Implied/realized volatility:	Current 3m implied volatility / current realized 3m volatility

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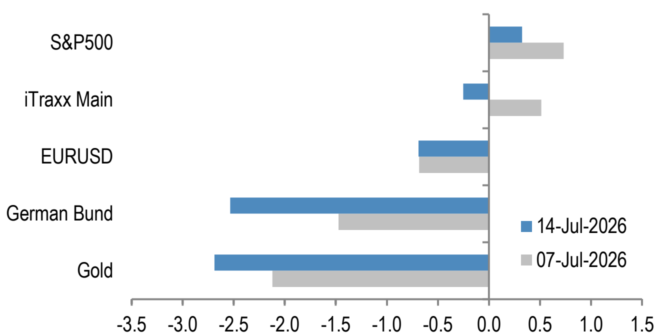
Chart A11b: Option skew monitor

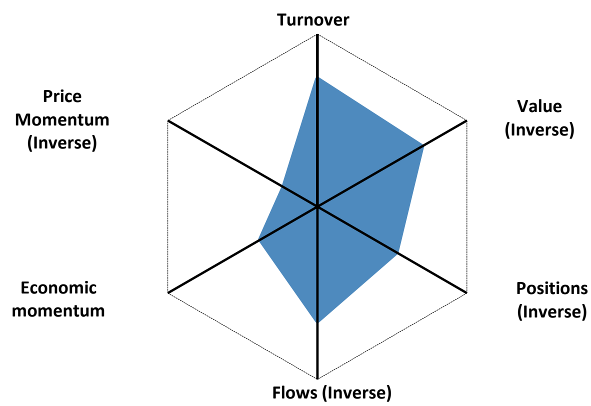
Skew is the difference between the implied volatility of out-of-the-money (OTM) call options and put options. A positive skew implies more demand for calls than puts and a negative skew, higher demand for puts than calls. It can therefore be seen as an indicator of risk perception in that a highly negative skew in equities is indicative of a bearish view. The chart shows z-score of the skew, i.e. the skew minus a rolling 2-year avg skew divided by a rolling two-year standard deviation of the skew. A negative skew on iTraxx Main means investors favour buying protection, i.e. a short risk position. A positive skew for the Bund reflects a long duration view, also a short risk position.



Source: J.P. Morgan Flows & Liquidity.

Chart A11c: Equity market health map

Each of the six axes corresponds to a key indicator for markets. The position of the blue line on each axis shows how far the current observation is from the extremes at either end of the scale. Overall, the larger the blue area within the hexagon, the better the outlook for the S&P500. All variables are expressed as the percentile of the distribution that the observation falls into. I.e. a reading in the middle of the axis means that the observation falls exactly at the median of all historical observations. **Turnover**: Percentile rank of the 21-day rolling S&P500 turnover since 2007. **Value**: Inverse of percentile rank of our quarterly S&P 500 Equity Risk Premium (ERP) estimate since 1957 as described in "A fair value model for US Bonds, Credit and Equities", Panigirtzoglou and Loeys, Jun 2005. **Positions**: Percentile rank of the difference between government bonds and equity positioning from the cross-asset positioning monitor on the front page of our publication since 2016. **Flows**: Inverse of percentile rank of the difference between equity and bond fund flows from the cross-asset fund flow monitor on the front page of our publication since 2007. **Economic momentum**: Percentile rank of the 2-month change in the global manufacturing PMI since 1998, see "REVISITING: Using the Global PMI as trading signal", Nikolaos Panigirtzoglou, Jan 2012). **Price momentum**: Inverse of the percentile rank of the z-score of the difference in the momentum signal of S&P500 vs. 10y USTs since 2016 (based on the average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 in the Appendix). China onshore A-share ETFs have been excluded. Last obs. as of 24th July.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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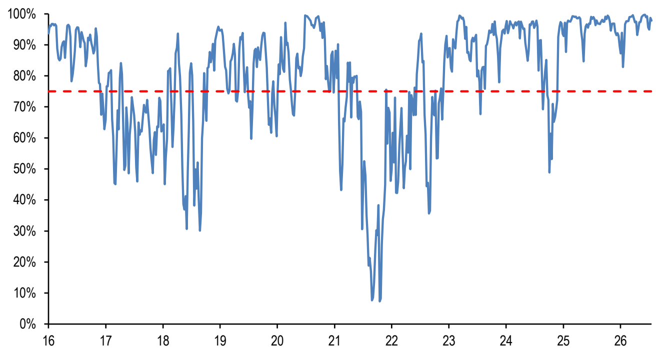
Chart A11d: Implied probability of the S&P500 rising over the next six months with a cutoff set at 75%



Source: J.P. Morgan Flows & Liquidity.

Chart A11e: Logistic regression model for direction of the S&P 500 over a 6 month horizon & Ex-ante probability of an up-move from the logistic regression algorithm vs. the ex-post realised 6-month return in the S&P 500

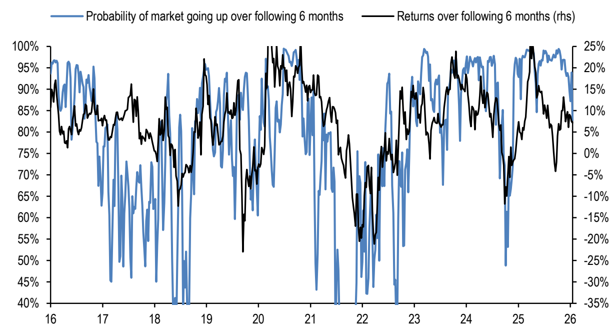
In %.

Dependent Variable: S&500 UP (1) or DOWN (0) over the next six months
Method: ML - Binary Logit (Newton-Raphson / Marquardt steps)

Sample (adjusted): 1/08/2016 12/20/2024
Included observations: 468 after adjustments

Coefficient covariance computed using observed Hessian

Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	3.42	0.84	4.08	0.00
TURNOVER	2.97	0.72	4.15	0.00
VALUE	-3.47	0.90	-3.88	0.00
POSITIONS	-0.77	0.55	-1.39	0.16
FLOWS	-1.79	0.53	-3.38	0.00
ECON_MOMENTUM	3.35	0.62	5.42	0.00
MOMENTUM	-1.36	0.62	-2.19	0.03



Source: J.P. Morgan Flows & Liquidity.

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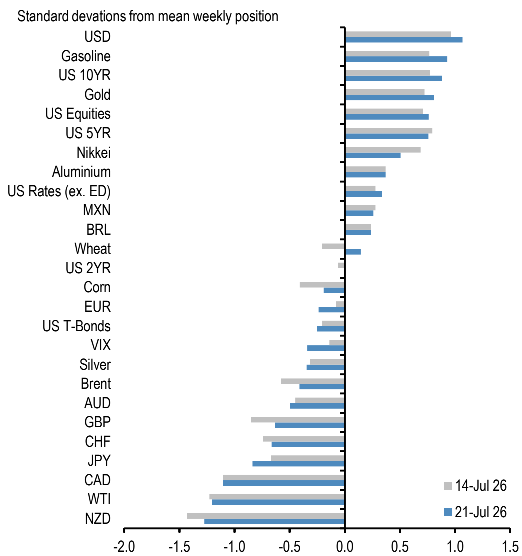
Spec position monitor

Chart A12: Weekly Spec Position Monitor

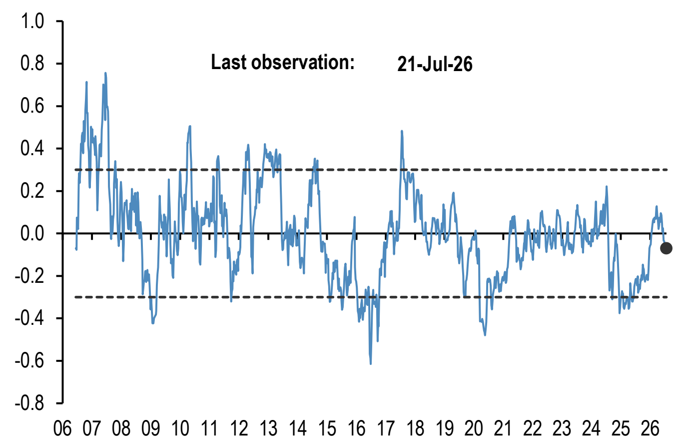
Net spec positions are proxied by the number of long contracts minus the number of short contracts using the speculative category of the Commitments of Traders reports (as reported by CFTC). To proxy for speculative investors for equity and US Treasury bond futures positions we use Asset managers and leveraged funds (see Chart A13), whereas for other assets we use the legacy Non-Commercial category. This net position is then converted to a dollar amount by multiplying by the contract size and then the corresponding futures price. We then scale the net positions by open interest. The chart shows the z-score of these net positions. US rates is a duration-weighted composite of the individual UST futures contracts excluding the Eurodollar contract.



Source: Bloomberg Finance L.P., CFTC, J.P. Morgan Flows & Liquidity.

Chart A14: Spec position indicator on Risky vs. Safe currencies

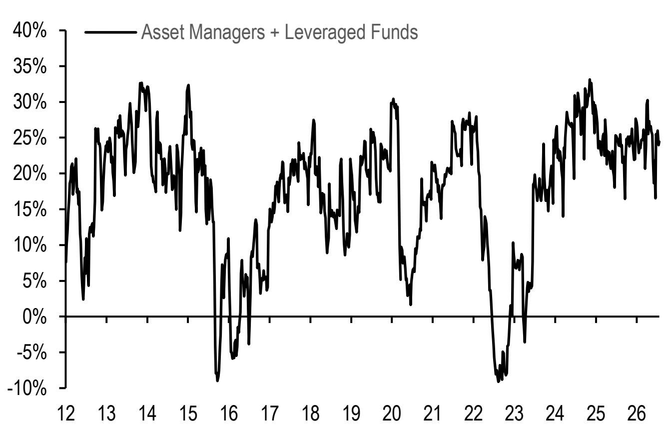
Difference between net spec positions on risky & safe currencies. Net spec position is calculated in USD across 5 'risky' and 3 'safe' currencies (safe currencies also include Gold). These positions are then scaled by open interest and we take an average of 'risky' and 'safe' assets to create two series. The chart is then simply the difference between the"risky" and "safe" series. The final series shown in the chart below is demeaned using data since 2006. The risky currencies are: AUD, NZD,CAD, MXN and BRL. The safe currencies are: JPY, CHF and Gold.



Source: Bloomberg Finance L.P., CFTC, J.P. Morgan Flows & Liquidity.

Chart A13: Positions in US equity futures by Asset managers and Leveraged funds

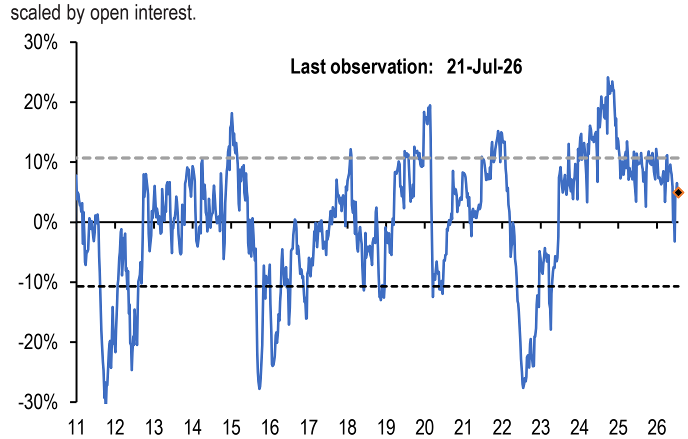
CFTC positions in US equity futures by Leveraged funds and Asset managers (as a % of open interest). It is an aggregate of the S&P500, DowJones, NASDAQ and their Mini futures contracts.



Source: CFTC, Bloomberg Finance L.P. and J.P. Morgan Flows & Liquidity.

Chart A15: Spec position indicator on US equity futures vs. intermediate sector UST futures

Difference between net spec positions on US equity futures vs. intermediate sector UST futures. This indicator is derived by the difference between total CFTC positions in US equity futures by Asset managers + Leveraged Funds scaled by open interest minus the Asset managers + Leveraged Funds spec position on intermediate sector UST futures (i.e. all UST futures duration weighted ex ED and ex 2Y UST futures) also scaled by open interest.



Source: CFTC, Bloomberg Finance L.P. and J.P. Morgan Flows & Liquidity.

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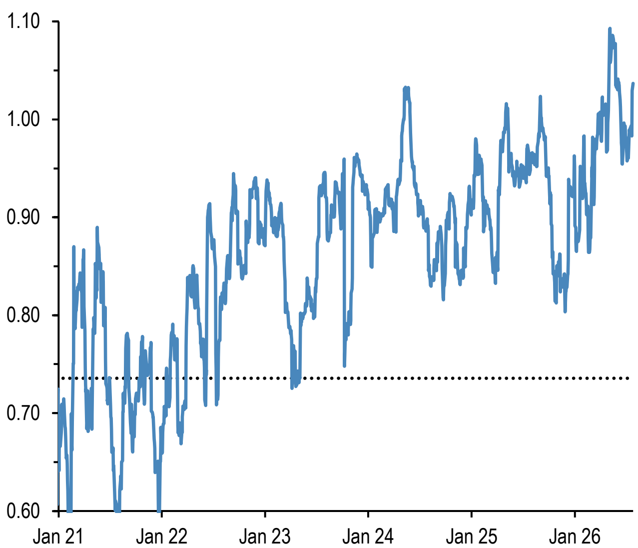
Mutual fund and hedge fund betas

Chart A16: 21-day rolling beta of 20 biggest active US bond mutual fund managers with respect to the US Agg Bond Index

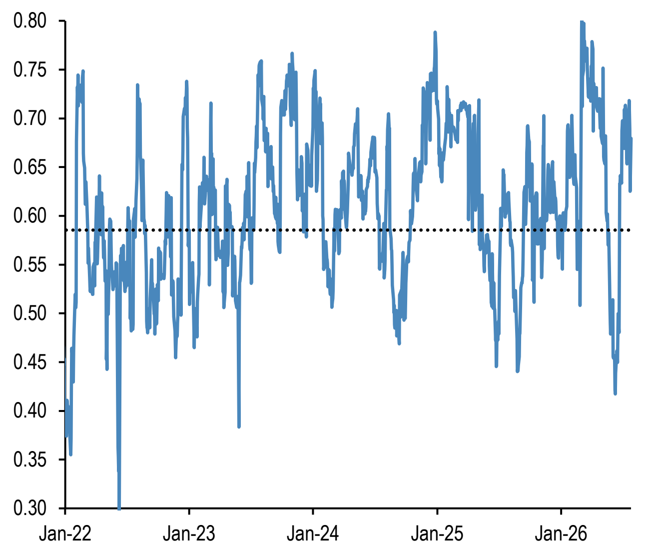
The dotted line shows the average beta since 2013.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A17: 21-day rolling beta of 20 biggest active Euro bond mutual fund managers with respect to the Euro Agg Bond Index

The dotted line shows the average beta since 2013.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A18: Performance of various types of investors

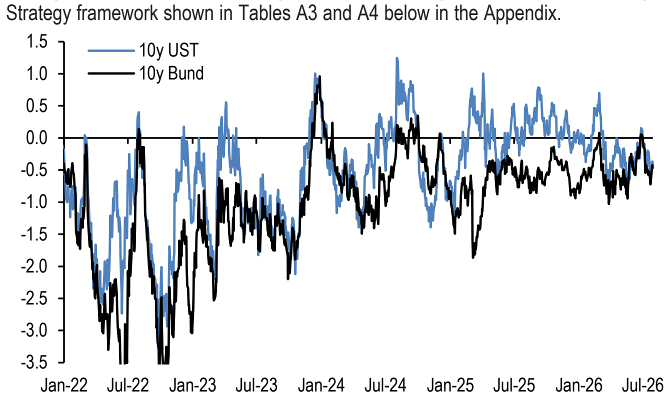
The table depicts the performance of various types of investors in % as of 27th July 2026.

Date	2020	2021	2022	2023	2024	2025	2026
Investors							
Equity L/S	13.4%	9.3%	-7.0%	11.3%	13.0%	15.5%	9.1%
Macro ex-CTAs	4.9%	4.3%	11.5%	1.3%	6.0%	10.2%	3.9%
CTAs	1.2%	8.8%	14.7%	-2.5%	2.9%	-1.1%	7.1%
Risk Parity Funds	3.5%	4.7%	-18.6%	6.0%	5.0%	17.3%	5.6%
US Balanced MFs	13.2%	14.4%	-13.0%	13.8%	11.4%	14.4%	6.3%
Benchmark							
MSCI AC World	16.3%	16.4%	-18.4%	22.2%	17.5%	22.3%	10.1%
Barclays Global Agg	5.6%	-2.5%	-11.2%	7.1%	3.4%	4.9%	0.3%
S&P Riskparity Vol 10	11.5%	12.8%	-16.2%	15.0%	5.4%	15.9%	13.8%
60 US Equity : 40 US Bonds	13.3%	14.8%	-15.4%	18.6%	16.4%	12.7%	5.7%

Source: Bloomberg Finance L.P., HFR, Pivotal Path, J.P. Morgan Flows & Liquidity.

Chart A19: Momentum signals for 10Y UST and 10Y Bunds

Average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 below in the Appendix.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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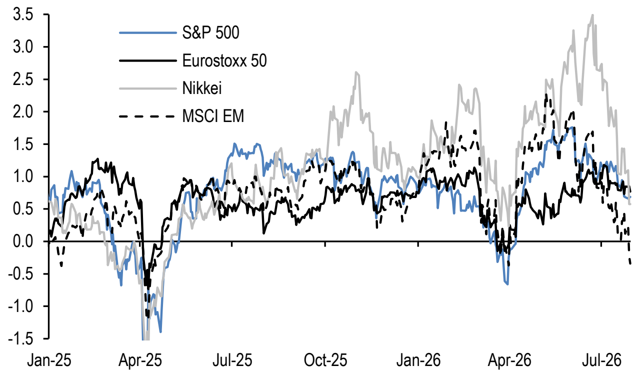
Chart A20: Momentum signals for S&P500

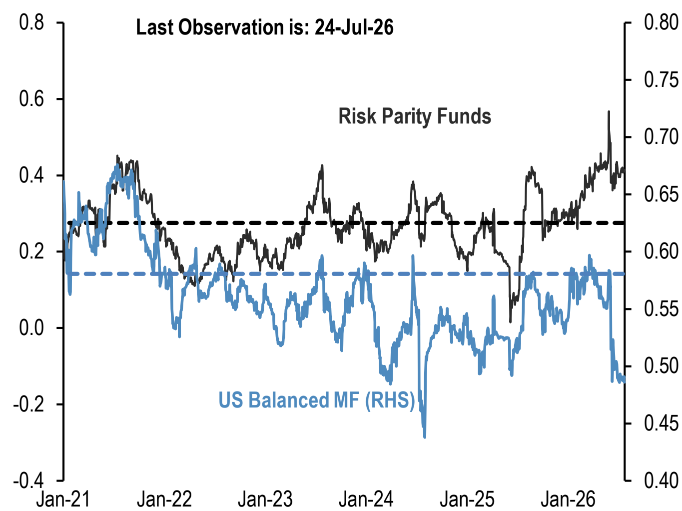
Average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 below in the Appendix.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A21: Equity beta of US Balanced Mutual funds and Risk Parity funds

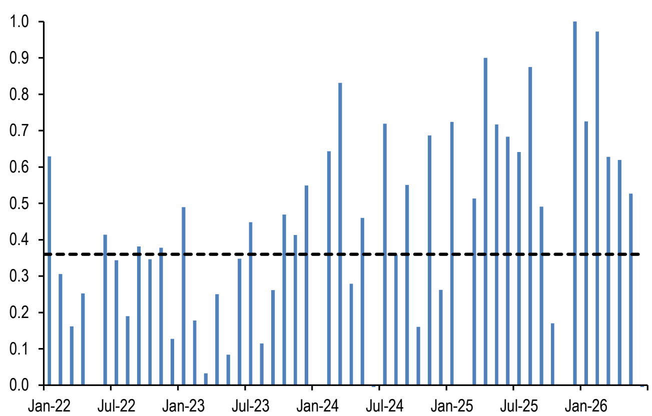
Rolling 42-day equity beta based on a bivariate regression of the daily returns of our Balanced Mutual fund and Risk Parity fund return indices to the daily returns of the S&P 500 and BarCap US Agg indices. Given that these funds invest in both equities and bonds we believe that the bivariate regression will be more suitable for these funds. Our risk parity index consists of 40 daily reporting Risk Parity funds. Our Balanced Mutual fund index includes the top 20 US-based active funds by assets and that have existed since 2006. Our Balanced Mutual fund index has a total AUM of $700bn, which is around half of the total AUM of $1.5tr of US based Balanced funds which we believe to be a good proxy of the overall industry It excludes tracker funds and funds with a low tracking error. Dotted lines are average since 2015.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A22: Equity beta of monthly reporting Equity Long/Short hedge funds

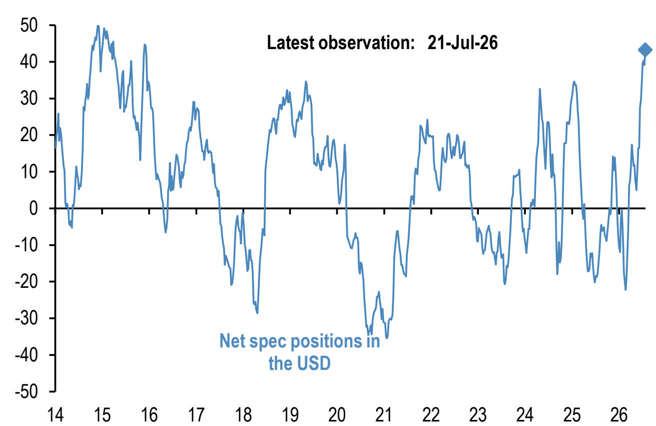
Proxied by the ratio of the monthly performance of Pivotal Path Asset-Weighted Equity Diversified hedge fund index divided by the monthly performance of MSCI AC World Index. Latest obs. is June 26 based on provisional data.



Source: Bloomberg Finance L.P., Pivotal Path, J.P. Morgan Flows & Liquidity.

Chart A23: USD exposure of currency hedge funds

The net spec position in the USD as reported by the CFTC. Spec is the non-commercial category from the CFTC.



Source: CFTC, Barclays, Datastream, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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CTAs – Trend following investors' momentum indicators

Table A3: Simple return momentum trading rules across various commodities

Optimal look-back period of each momentum strategy combined with a mean reversion indicator that turns signal neutral when momentum z-score more than 1.5 standard deviations above or below mean, and a filter that turns neutral when the z-score is low (below 0.05 and above -0.05) to avoid excessive trading. Look-backs, current signals and z-scores are shown for shorter-term and longer-term momentum separately, along with performance of a combined signal. Annualised return, volatility and information ratio of the signal; current signal; and z-score of the current return over the relevant look-back period; data from 1999 onward.

		Lookback (moving avg, days)	Annualized return (%)	Vol (%)	IR	Current signal	Time since last change (days)	Z-score	% Change of return index from its moving average
WTI	short	21	6.2	22.7	0.27	1	1	0.5	3.0%
	long	462				1	1	1.1	32.0%
Brent	short	84	4.3	21.5	0.20	-1	1	-0.6	-7.3%
	long	504				1	1	1.2	33.0%
Unleaded gas	short	105	3.5	23.2	0.15	1	44	0.7	10.3%
	long	483				0	21	2.2	59.1%
Heat Oil	short	63	4.7	22.0	0.21	1	1	1.3	13.7%
	long	483				0	105	3.0	93.7%
Gasoil	short	63	8.2	21.2	0.39	1	0	1.4	14.8%
	long	378				0	105	3.1	90.4%
Nat gas	short	105	13.6	35.9	0.38	-1	14	-0.8	-14.1%
	long	315				-1	126	-0.8	-24.9%
EU emission allowances	short	42	10.2	28.8	0.35	1	6	0.2	2.1%
	long	483				1	34	0.2	7.5%
1m fwd TTF Nat gas	short	21	36.9	31.4	1.18	1	1	1.1	10.6%
	long	294				1	1	1.2	55.5%
Gold	short	21	3.1	10.6	0.29	-1	0	-0.3	-0.9%
	long	483				1	37	0.4	5.5%
Silver	short	10	6.1	19.7	0.31	-1	0	-0.3	-0.9%
	long	462				1	28	0.4	9.6%
Palladium	short	42	12.0	23.1	0.52	0	0	0.0	-0.4%
	long	273				-1	50	-0.6	-15.0%
Platinum	short	105	4.3	18.6	0.23	0	28	-1.6	-15.1%
	long	273				-1	28	-0.8	-12.3%
Aluminium	short	105	5.7	15.2	0.38	-1	26	-1.0	-7.8%
	long	357				1	31	0.6	8.0%
Copper	short	147	7.3	17.3	0.42	1	118	0.3	3.7%
	long	399				1	124	1.0	21.7%
Lead	short	126	0.8	19.2	0.04	-1	36	-0.4	-4.1%
	long	357				-1	137	-0.4	-9.0%
Nickel	short	42	11.8	23.0	0.51	-1	38	-0.3	-2.6%
	long	336				0	0	0.0	0.7%
Zinc	short	126	7.9	19.6	0.40	1	86	0.4	4.9%
	long	399				1	216	0.8	17.8%
Wheat	short	168	4.3	23.3	0.18	1	19	0.8	10.0%
	long	294				1	19	0.7	11.5%
Kansas wheat	short	147	8.6	20.8	0.41	1	2	1.2	13.8%
	long	483				1	16	0.7	14.3%
Corn	short	63	7.6	16.6	0.46	1	4	0.3	2.0%
	long	399				-1	60	-0.3	-5.5%
Soybeans	short	42	4.6	14.5	0.32	1	16	0.8	4.0%
	long	231				1	16	0.4	4.9%
Cotton	short	168	5.3	18.5	0.29	1	33	0.4	5.2%
	long	483				1	1	0.1	1.8%
Sugar	short	63	6.4	21.6	0.30	-1	4	-0.2	-2.0%
	long	252				-1	53	-0.3	-5.4%
Coffee	short	63	7.9	24.0	0.33	0	1	2.2	19.5%
	long	273				1	2	0.5	9.5%
Cocoa*		10	5.5	31.3	0.18	-1	8	-1.2	-4.7%

* For cocoa, uses only short-term momentum and a z-score threshold of 3 rather than 1.5 as for other contracts.

Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Table A4: Simple return momentum trading rules across international equity indices, bond futures and FX

Optimal look-back period of each momentum strategy combined with a mean reversion indicator that turns signal neutral when momentum z-score more than 1.5 standard deviations above or below mean, and a filter that turns neutral when the z-score is low (below 0.05 and above -0.05) to avoid excessive trading. Look-backs, current signals and z-scores are shown for shorter-term and longer-term momentum separately, along with performance of a combined signal. Annualised return, volatility and information ratio of the signal; current signal; and z-score of the current return over the relevant look-back period; data from 1999 onward.

		Lookback (moving avg, days)	Annualized return (%)	Vol (%)	IR	Current signal	Time since last change (days)	Z-score	% Change of return index from its moving average
S&P 500	short	84	7.6	12.3	0.62	1	0	0.3	1.7%
	long	315				1	0	1.0	10.5%
Nasdaq 100	short	84	10.0	16.2	0.62	-1	0	-0.3	-2.3%
	long	462				1	0	0.9	17.4%
Nikkei	short	63	3.9	13.3	0.29	-1	0	-0.9	-5.2%
	long	420				0	0	2.0	31.2%
FTSE 100	short	168	5.3	12.1	0.44	1	0	1.0	6.4%
	long	504				0	0	1.8	20.9%
Eurostoxx 50	short	168	4.9	16.6	0.30	1	0	0.8	6.6%
	long	294				1	0	0.9	11.0%
MSCI EM	short	42	12.7	11.5	1.11	-1	0	-1.4	-7.0%
	long	357				1	0	0.9	14.7%
HYG credit	short	168	0.4	3.4	0.12	-1	0	-0.2	-0.6%
	long	273				-1	0	-0.2	-0.6%
LQD credit	short	10	2.6	6.5	0.40	-1	0	-0.2	-0.2%
	long	189				-1	0	-0.2	-0.8%
2Y USTs	short	252	0.9	1.1	0.79	-1	0	-1.0	-1.1%
	long	462				-1	0	-0.6	-1.2%
5Y USTs	short	252	1.5	2.8	0.55	-1	0	-0.8	-2.0%
	long	420				-1	0	-0.5	-1.8%
10Y USTs	short	42	2.0	3.6	0.55	-1	0	-0.4	-0.5%
	long	441				-1	0	-0.4	-1.8%
2Y Schatz	short	189	0.6	0.8	0.81	-1	0	-0.8	-0.6%
	long	504				-1	0	-1.3	-2.2%
5y Bobl	short	63	1.3	1.9	0.71	-1	0	-0.4	-0.3%
	long	483				-1	0	-0.8	-2.8%
10y Bund	short	105	2.2	3.4	0.63	-1	0	-0.2	-0.5%
	long	462				-1	0	-0.6	-3.1%
10Y JGB	short	126	0.9	2.3	0.38	-1	0	-1.0	-1.1%
	long	273				0	0	-1.9	-2.9%
10Y Gilts	short	42	2.3	4.1	0.56	-1	0	-0.5	-0.8%
	long	399				-1	0	-0.8	-4.0%
10y OAT vs Bund	short	189	0.4	1.5	0.25	-1	0	-0.1	-0.1%
	long	294				1	0	0.3	0.3%
10y BTP vs. Bund	short	84	2.2	6.3	0.35	0	0	0.0	-0.1%
	long	273				0	0	0.0	0.2%
Euro	short	42	3.4	6.4	0.53	-1	0	-0.5	-1.1%
	long	273				-1	0	-0.9	-5.0%
Yen	short	21	3.7	6.3	0.59	-1	0	-0.7	-1.1%
	long	462				0	0	-1.6	-12.9%
Sterling	short	168	1.4	7.1	0.20	-1	0	-0.5	-2.3%
	long	294				-1	0	-0.6	-3.1%
AUD	short	42	3.8	7.8	0.49	-1	0	-0.2	-0.6%
	long	420				1	0	0.2	1.8%
CAD	short	168	1.4	5.9	0.25	-1	0	-1.0	-3.8%
	long	504				-1	0	-1.0	-6.9%
G10 FX	short	42	2.2	5.2	0.42	1	0	0.5	0.9%
	long	189				1	0	0.7	2.6%
EMCI	short	42	2.6	5.0	0.52	-1	0	-0.4	-0.7%
	long	189				1	0	0.2	0.9%

Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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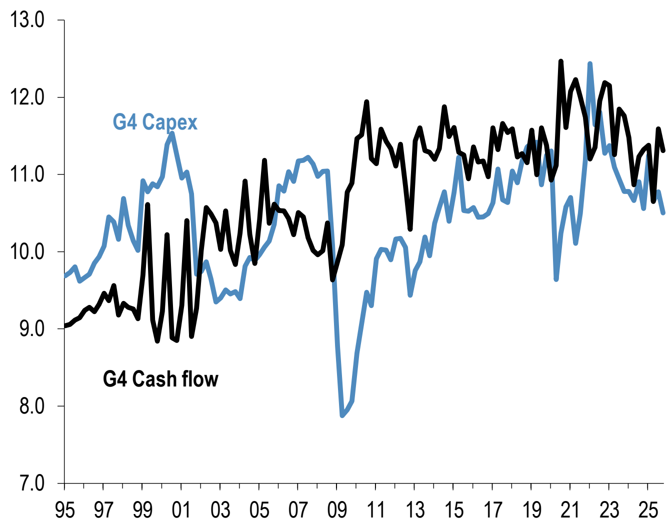
Corporate Activity

Chart A24: G4 non-financial corporate capex and cash flow as % of GDP

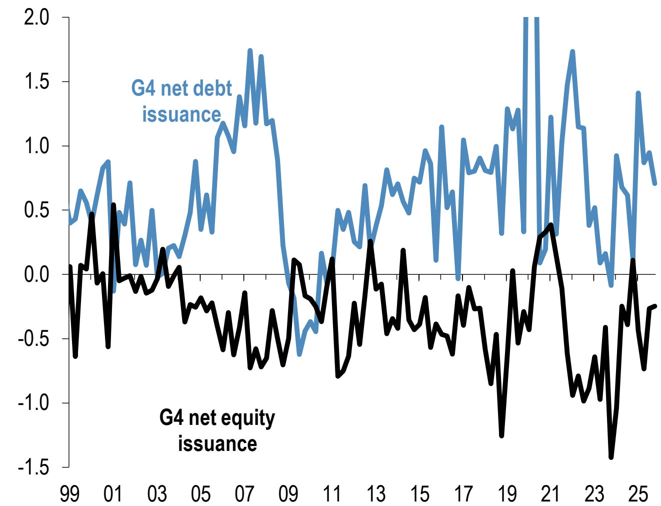
% of GDP, G4 includes the US, the UK, the Euro area and Japan. Last observation as of Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A25: G4 non-financial corporate sector net debt and equity issuance

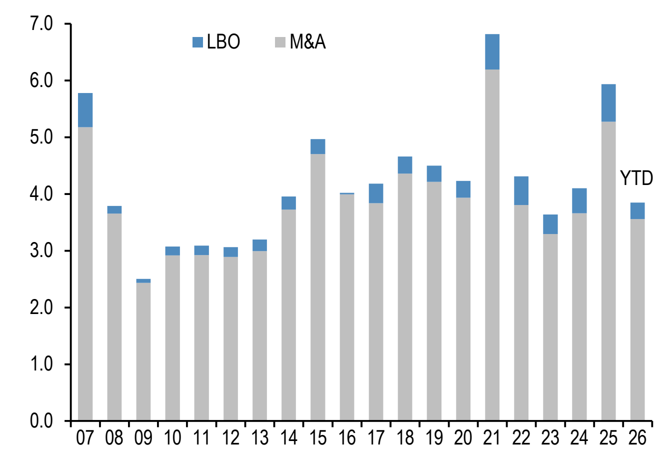
$tr per quarter, G4 includes the US, the UK, the Euro area and Japan. Last observation as of Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A26: Global M&A and LBO

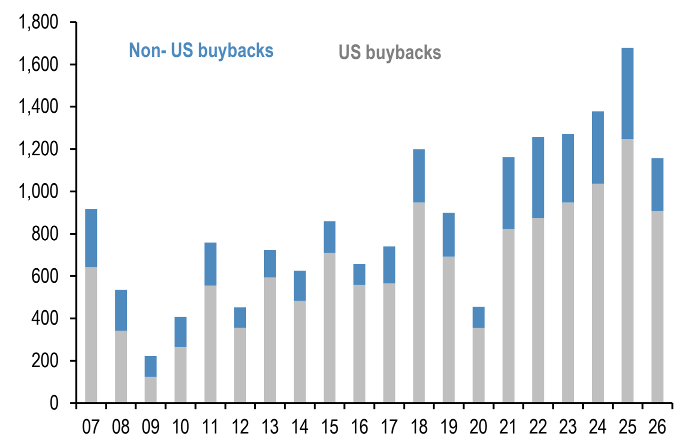
$tr. M&A and LBOs are announced.



Source: Dealogic, J.P. Morgan Flows & Liquidity.

Chart A27: US and non-US share buyback

$bn, are as of July 26. Buybacks are as announced.



Source: Bloomberg Finance L.P., LSEG, J.P. Morgan Flows & Liquidity.

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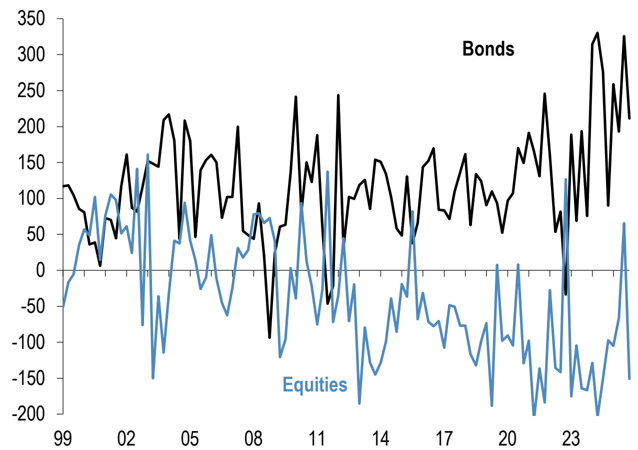
Pension fund and insurance company flows

Chart A28: G4 pension funds and insurance companies equity and bond flows

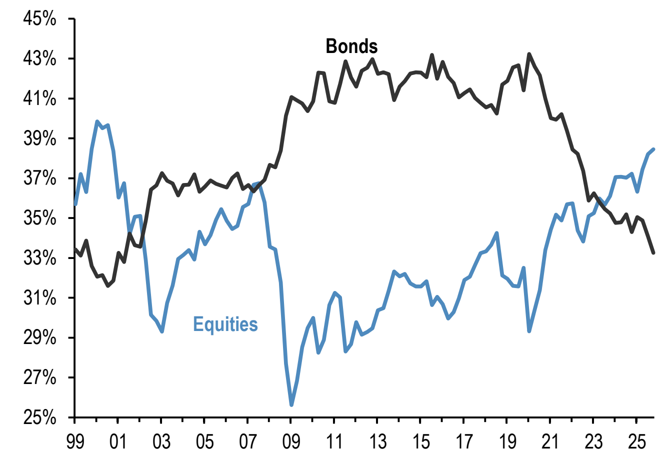
Equity and bond buying in $bn per quarter. G4 includes the US, the UK, Euro area and Japan. Last observation is Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A29: G4 pension funds and insurance companies equity and bond levels

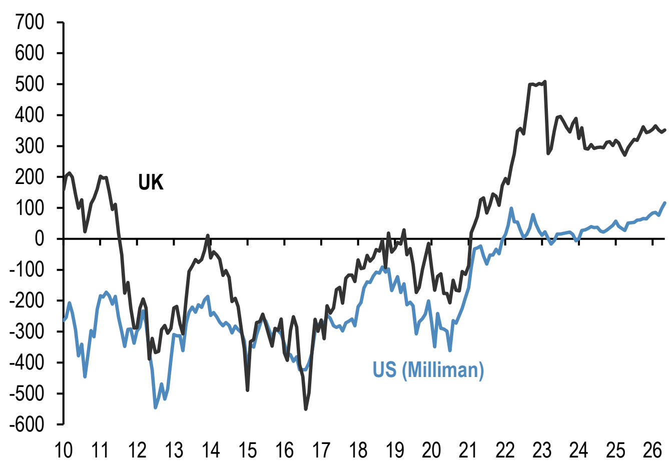
Equity and bond as % of total assets per quarter. G4 includes the US, the UK, Euro area and Japan. Last observation is Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A30: Pension fund deficits

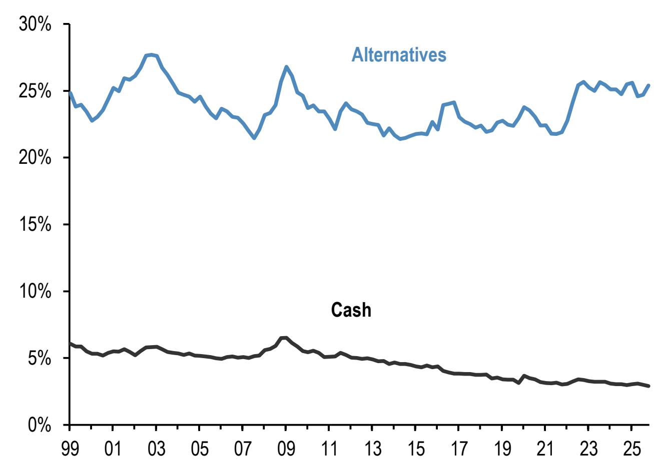
US$bn. For US, funded status of the 100 largest corporate defined benefit pension plans, from Milliman. For UK, funded status of the defined benefit schemes eligible for entry to the Pension Protection Fund, converted to US$ at today's exchange rates. Last obs. is May'26 for US & UK.



Source: Milliman, UK Pension Protection Fund, J.P. Morgan Flows & Liquidity.

Chart A31: G4 pension funds and insurance companies cash and alternatives levels

Cash and alternative investments as % of total assets per quarter. G4 includes the US, the UK, Euro area and Japan. Last observation is Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

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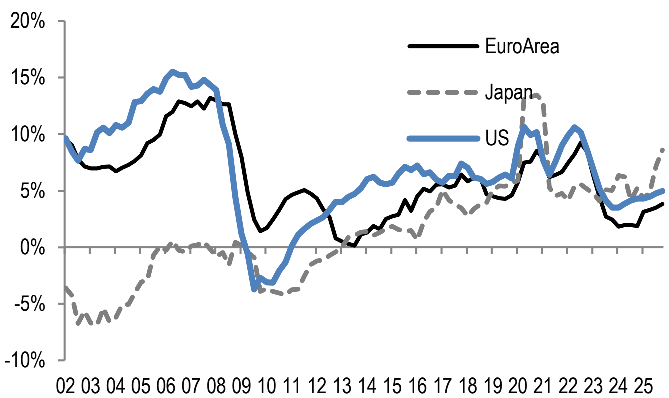
Credit Creation

Chart A32: Credit creation in the US, Japan and Euro area

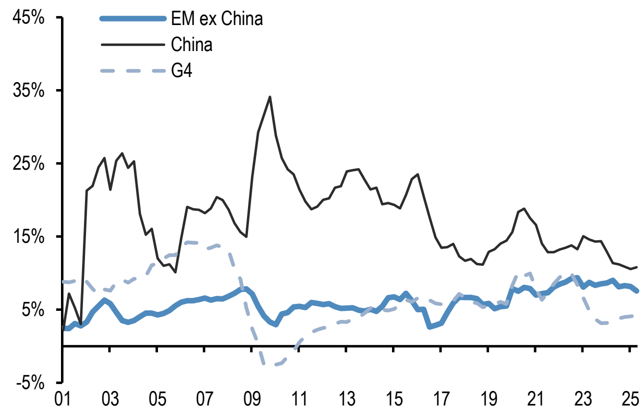
Rolling sum of 4-quarter credit creation as % of GDP. Credit creation includes both bank loans as well as net debt issuance by non-financial corporations and households. Last obs. is Q4'25 for Japan, Euro Area and US.



Source: Fed, ECB, BoJ, Bloomberg Finance L.P., and J.P. Morgan Flows & Liquidity.

Chart A33: Credit creation in EM

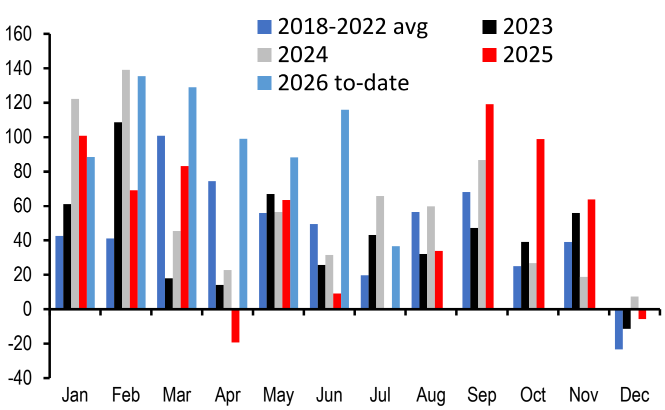
Rolling sum of 4-quarter credit creation as % of GDP. Credit creation includes both bank loans as well as net debt issuance by non-financial corporations and households. The chart uses the FX rates as of the latest observation to calculate the credit creation at constant exchange rates. Last obs. is for Q2'25.



Source: G4 Central banks FoF, BIS, ICI, Barcap, Bloomberg Finance L.P., IMF, and J.P. Morgan Flows & Liquidity.

Chart A34: Monthly net issuance of US HG bonds

$bn. Up to July 23rd.



Source: Dealogic, J.P. Morgan Flows & Liquidity.

Table A5: Equity and Bond issuance

$bn, Equity supply and corporate announcements are based on announced deals, not completed. M&A is announced deal value and buybacks are announced transactions. Y/Y change is change in YTD announcements over the same period last year.

Equity Supply	17-Jul	4 wk avg	13 wk avg	y/y chng
Global IPOs	9.7	6.0	12.5	197%
Secondary Offerings	3.9	19.9	15.6	36%
Corporate announcements				
M&A - Global	71.1	108.2	141.8	45%
- US Target	27.6	49.5	79.7	74%
- Non-US Target	43.5	58.7	62.1	24%
Net bond issuance	**Feb-26**	**3 mth avg**	**YTD avg**	**y/y chng**
USD	8	144	179	-50%
Non-USD	162	129	277	-4%

Source: Bloomberg Finance L.P., Dealogic, Thomson Reuters, J.P. Morgan Flows & Liquidity.

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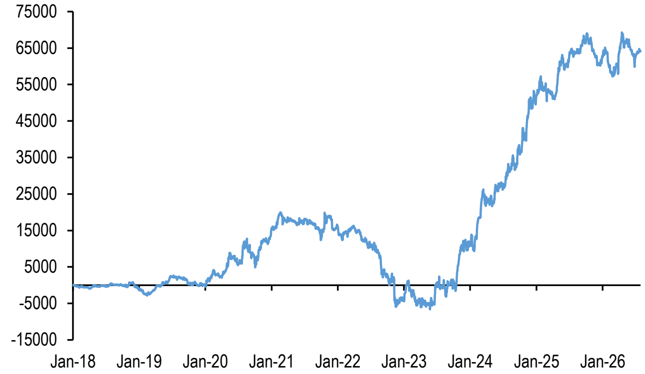
Bitcoin monitor

Chart A35: Our Bitcoin position proxy, based on open interest in CME Bitcoin futures contracts

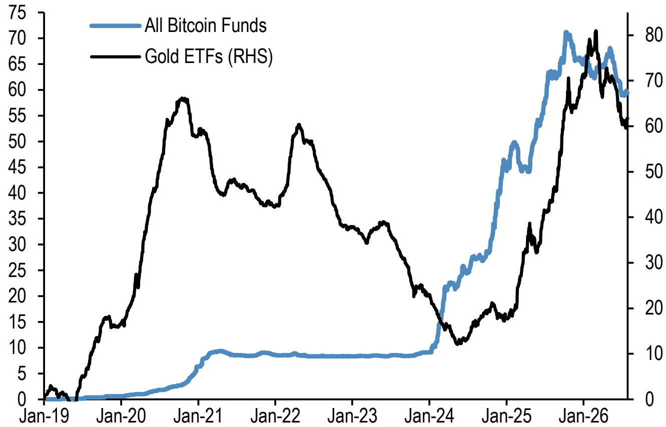
In number of contracts. Last obs. for 28th July 2026.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A36: Cumulative Flows in all Bitcoin funds and Gold ETF holdings

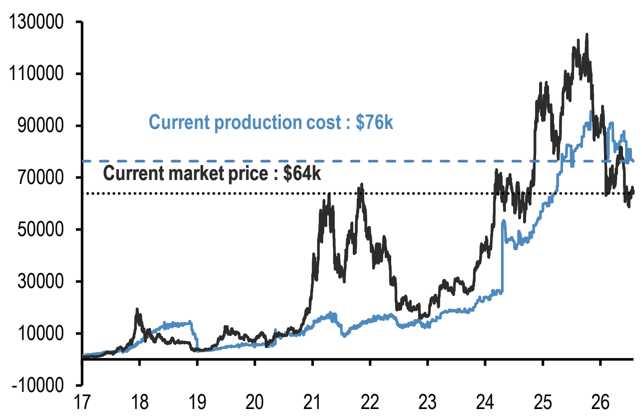
Both the y-axis in $bn.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A37: Ratio of bitcoin market price to production cost

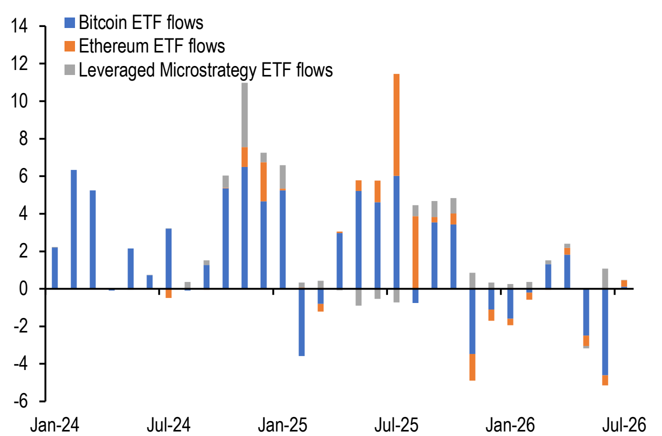
Based on the cost of production approach following Hayes (2018).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A38: Monthly flows into spot bitcoin ETFs, spot ethereum ETFs and leveraged MicroStrategy ETF

$bn.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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Japanese flows and positions

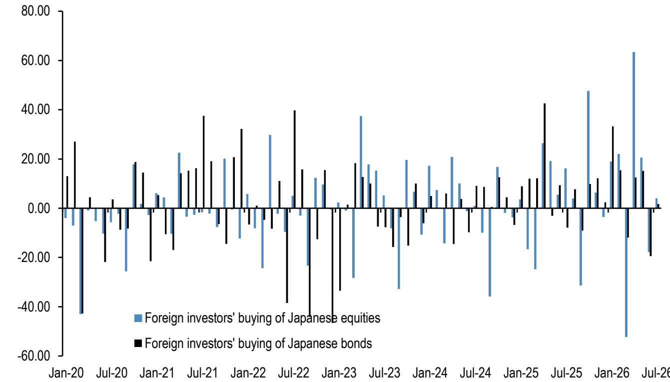
Chart A39: Tokyo Stock Exchange margin trading: total buys minus total sells

In bn of shares. Topix on right axis.

Source: Tokyo Stock Exchange, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A40: Monthly net purchases of Japanese bonds and Japanese equities by foreign residents

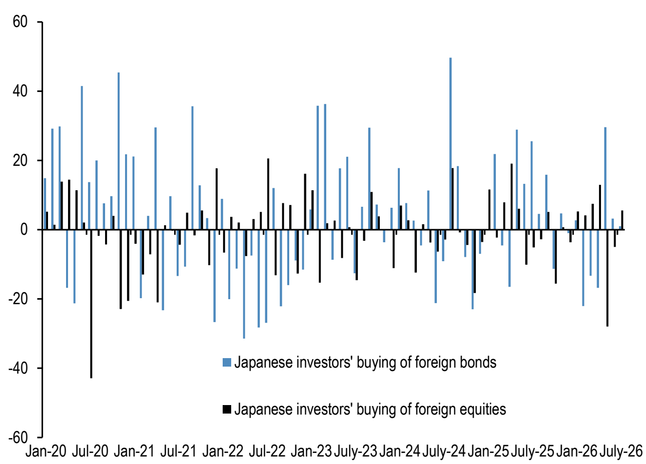
$bn, Last weekly obs. is for 17th July' 26.



Source: Japan MoF, Bloomberg Finance L.P., and J.P. Morgan Flows & Liquidity.

Chart A41: Monthly net purchases of foreign bonds and foreign equities by Japanese residents

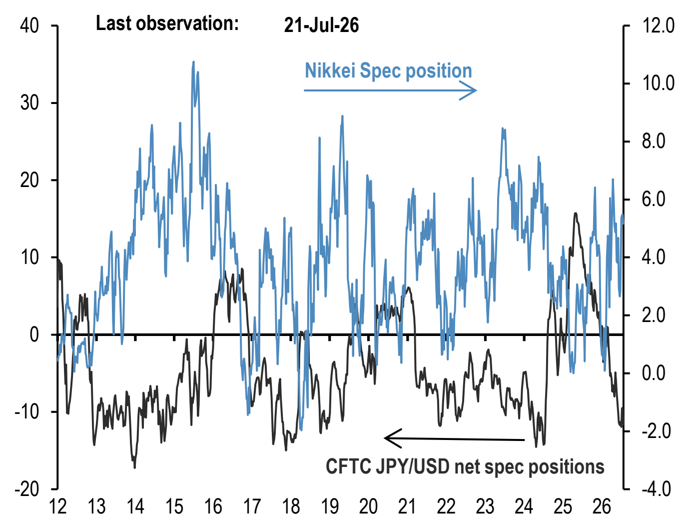
$bn, Last weekly obs. is for 17thJuly' 26.



Source: Japan MoF, Bloomberg Finance L.P., and J.P. Morgan Flows & Liquidity.

Chart A42: Overseas CFTC spec positions

CFTC spec positions are in $bn. For Nikkei we use CFTC positions in Nikkei futures (USD & JPY) by Leveraged funds and Asset managers.



Source: Bloomberg Finance L.P., CFTC, J.P. Morgan Flows & Liquidity.

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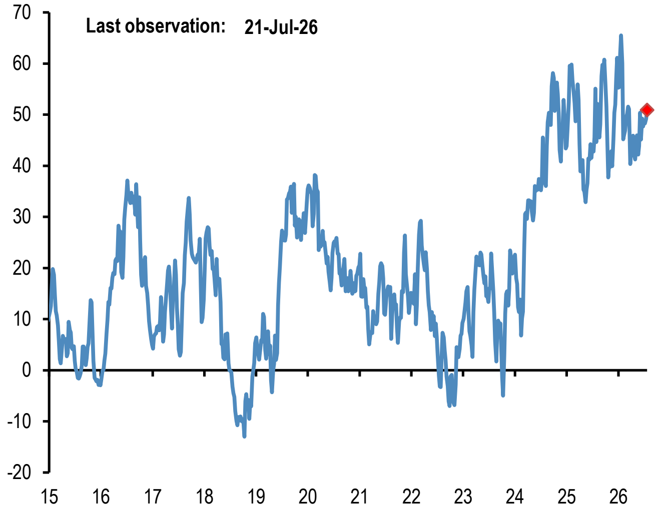
Commodity flows and positions

Chart A43: Gold spec positions

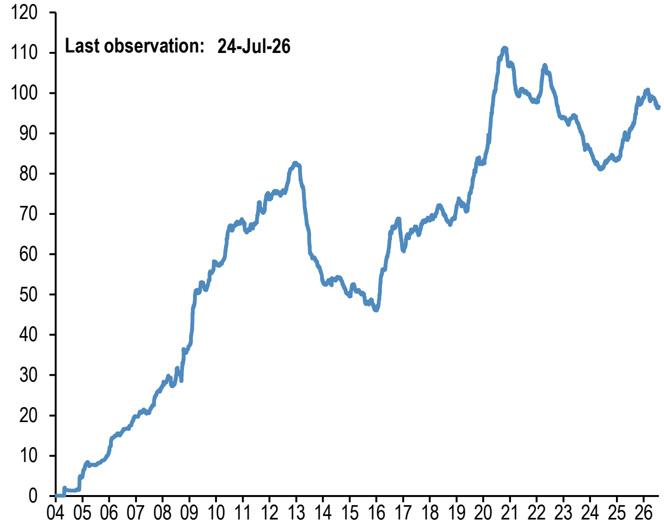
$bn. CFTC net long minus short position in futures for the Managed Money category.



Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A44: Gold ETFs

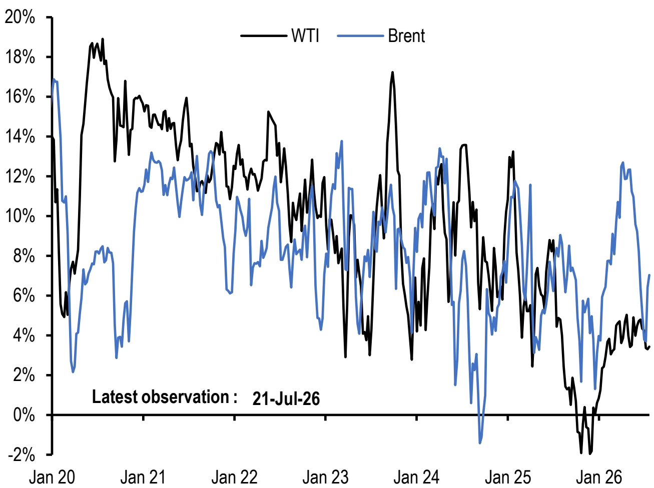
Mn troy oz. Physical gold held by all gold ETFs globally.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A45: Oil spec positions

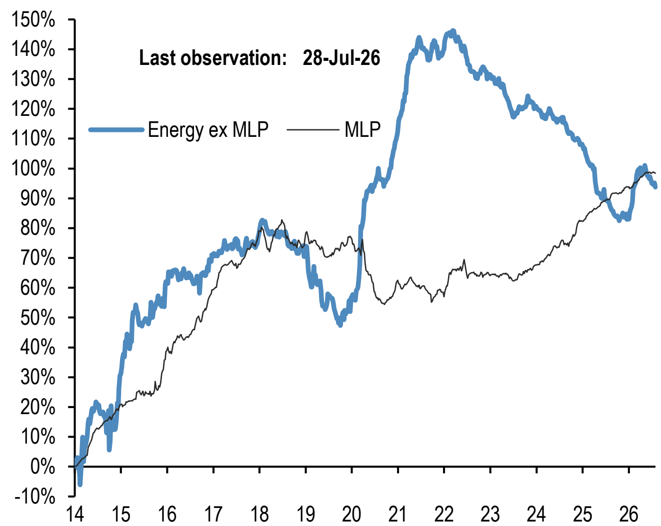
Net spec positions divided by open interest. CFTC futures positions for WTI and Brent are net long minus short for the Managed Money category.



Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A46: Energy ETF flows

Cumulative energy ETFs flows as a % of AUM. MLP refers to the Alerian MLP ETF.



Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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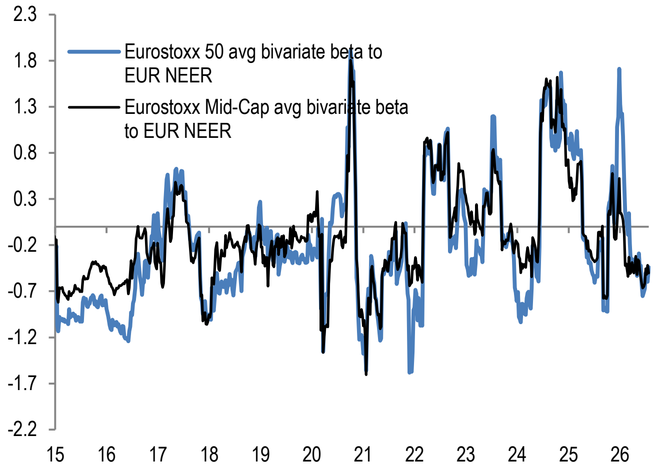
Corporate FX hedging proxies

Chart A47: Average beta of Eurostoxx 50 companies and Eurostoxx Mid-Cap to trade-weighted EUR

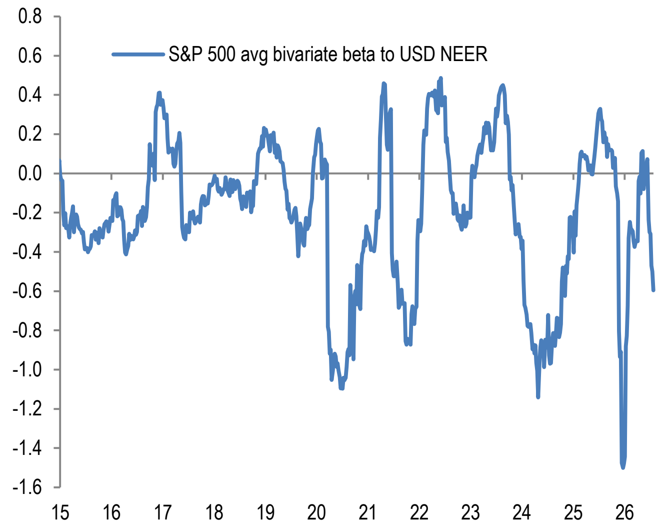
Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of individual stocks in the Eurostoxx 50 index to the weekly returns of the MSCI AC World and JPM EUR Nominal broad effective exchange rate (NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A48: Average beta of S&P500 companies to trade-weighted US dollar

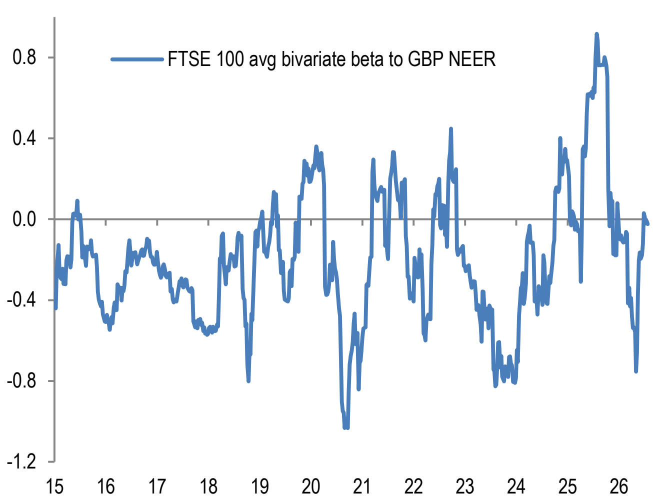
Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of stocks in the S&P500 index to the weekly returns of the MSCI AC World and JPM USD Nominal broad effective exchange rate(NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A49: Average beta of FTSE 100 companies to trade-weighted GBP

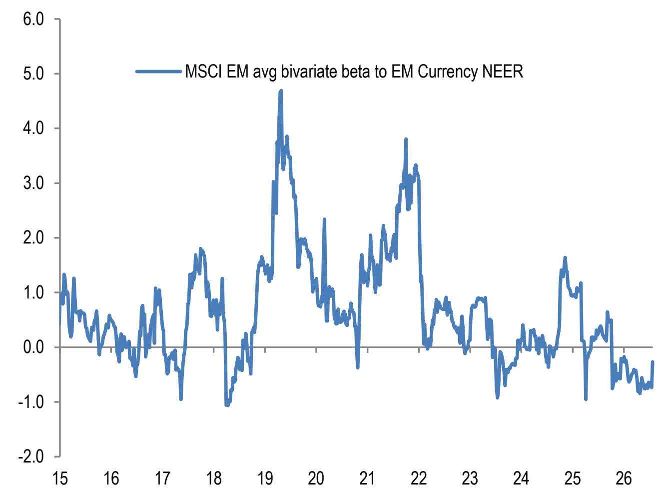
Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of individual stocks in the FTSE 100 index to the weekly returns of the MSCI AC World and JPM GBP Nominal broad effective exchange rate (NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A50: Average beta of MSCI EM companies to trade-weighted EM Currency Index

Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of individual stocks in the MSCI EM index to the weekly returns of the MSCI AC World and JPM EM Nominal broad effective exchange rate (NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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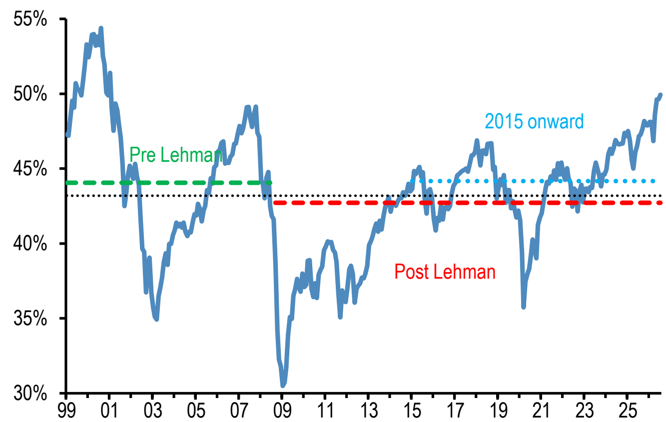
Non-Bank investors' implied allocations

Chart A51: Implied equity allocation by non-bank investors globally

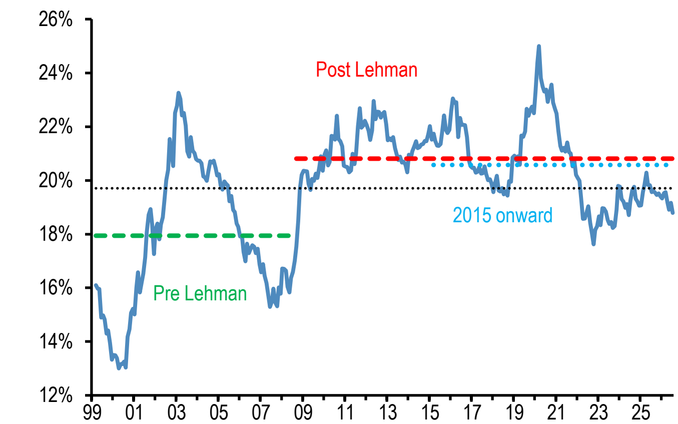
Global equities as % total holdings of equities/bonds/M2 by non-bank investors/ Commodities including gold held by private investors. Dotted lines are averages.



Source: Bloomberg Finance L.P., LSEG DataStream, J.P. Morgan Flows & Liquidity.

Chart A52: Implied bond allocation by non-bank investors globally

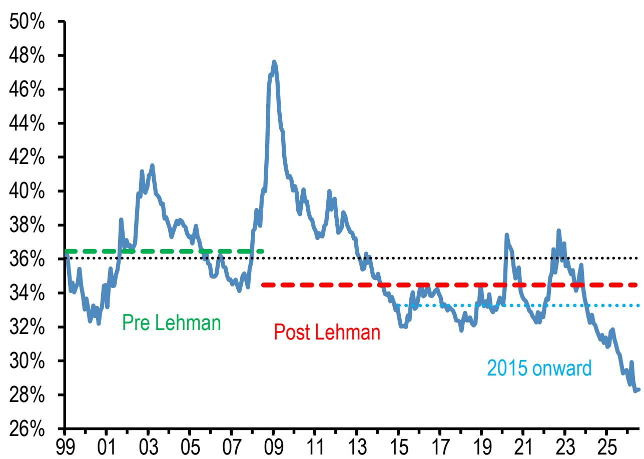
Global bonds as % total holdings of equities/bonds/M2 by non-bank investors/ Commodities including gold held by private investors. Dotted lines are averages.



Source: Bloomberg Finance L.P., LSEG DataStream, J.P. Morgan Flows & Liquidity.

Chart A53: Implied cash allocation by non-bank investors globally

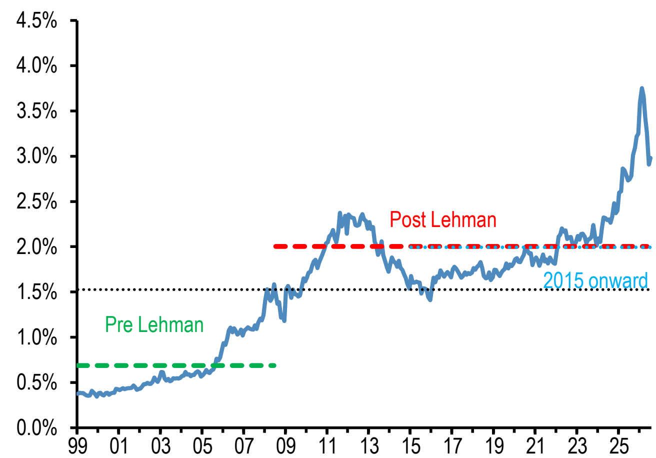
Global cash held by non-bank investors as % total holdings of equities/bonds/M2 by non-bank investors/Commodities including gold held by private investors. Dotted lines are averages.



Source: Bloomberg Finance L.P., LSEG DataStream, J.P. Morgan Flows & Liquidity.

Chart A54: Implied commodity allocation (including privately held gold) by non-bank investors globally

Proxied by the open interest of commodity futures ex gold and privately held gold as % of the stock of equities, bonds, cash held by non-bank investors globally and commodities including gold held by private investors.



Source: Bloomberg Finance L.P., LSEG DataStream, J.P. Morgan Flows & Liquidity.

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